

LCFC Confidential

AILZA/B/C (ZX10)

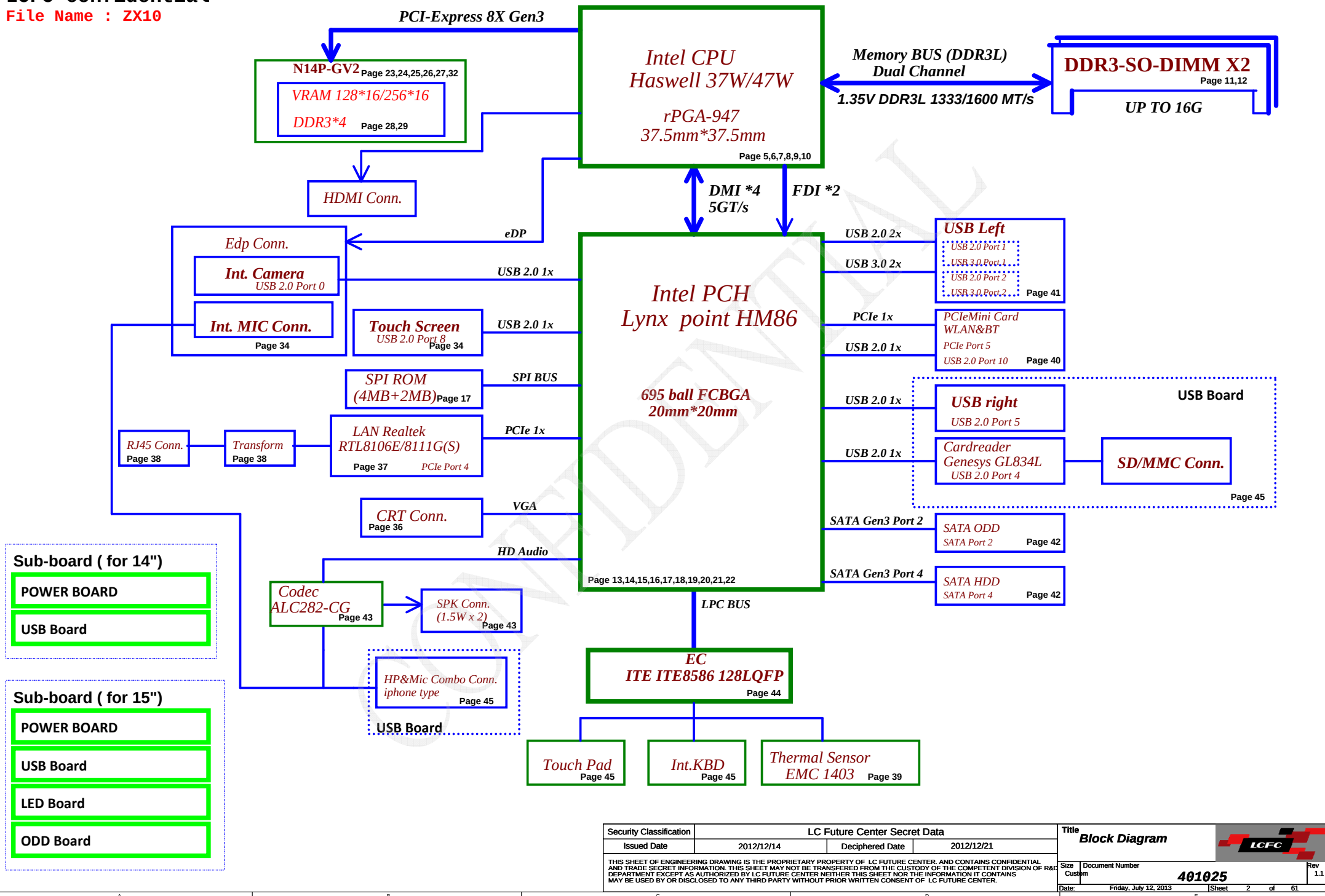
MB MA181 Schematics

Intel Haswell Processor with DDRIIL + Lynx Point HM86
nVIDIA N14P-GV2

2012-12-27

REV:1.1

Security Classification	LC Future Center Secret Data			Title	Cover Page		
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Voltage Rails (O --> Means ON , X --> Means OFF)

Power Plane	State	B+	+3VALW	+5VALW	+3V_PCH	+1.35V	+5VS +3VS +1.5VS +1.05VS +0.675VS +CPU_CORE +VGA_CORE +3.3VS_VGA +1.5VS_VGA +1.05VS_VGA
S0		0	0	0	0	0	0
S3		0	0	0	0	0	X
S3 Battery only		0	0	X	0	X	X
S5 S4/AC Only		0	0	0	X	X	X
S5 S4 Battery only		0	X	X	X	X	X
S5 S4 AC & Battery don't exist		X	X	X	X	X	X

SMBUS Control Table

	SOURCE	VGA	BATT	IT8586E	SODIMM	WLAN	Thermal Sensor	PCH	TP Module	charger	XDP
EC_SMB_CK1 EC_SMB_DA1	IT8586E +3VALW	X	V	V	X	X	X	X	X	V	X
EC_SMB_CK2 EC_SMB_DA2	IT8586E +3VS	V +3VS_VGA	X	V +3VS	X	X	V +3VS	V +3V_PCH	X	X	X
PCH_SMBCLK PCH_SMBDATA	PCH +3V_PCH	X	X	X	V +3VS	V +3VS	X	V +3V_PCH	V +3VS	X	V +3VS

EC SM Bus1 address

EC SM Bus2 address

PCH SM Bus address

Device	Address	Device	Address	Device	Address
Smart Battery	0001 011X b	Thermal Sensor EMC1403-2	1001_101xb	DDR DIMMA	0xA0
Charger	0001 0010 b	VGA	0x9E	DDR DIMMB	0xA2
		PCH	0x96	Wlan	Rsvd
				TP	0x2C for Synaptics 0x15 for ELAN vendor

STATE	SIGNAL	SLP_S1#	SLP_S3#	SLP_S4#	SLP_S5#	+VALW	+V	+VS	Clock
Full ON		HIGH	HIGH	HIGH	HIGH	ON	ON	ON	ON
S1(Power On Suspend)		LOW	HIGH	HIGH	HIGH	ON	ON	ON	LOW
S3 (Suspend to RAM)		LOW	LOW	HIGH	HIGH	ON	ON	OFF	OFF
S4 (Suspend to Disk)		LOW	LOW	LOW	HIGH	ON	OFF	OFF	OFF
S5 (Soft OFF)		LOW	LOW	LOW	LOW	ON	OFF	OFF	OFF

USB Port Table

USB 2.0	USB 3.0	Port	3 External USB Port
EHCI1	XHCI 1	0	Camera
		1	USB Port (Left Side)
		2	USB Port (Left Side)
		3	
		4	Cardreader
		5	USB Port (Right Side)
		6	
EHCI2	2	7	
		8	TOUCH PANEL
		9	
		10	Mini Card(WLAN)
		11	
		12	
		13	

PCIE PORT LIST

Port	Device
1	
2	
3	
4	LAN
5	WLAN
6	
7	
8	

BOM Structure Table

BOM Structure	BTO Item
AOAC@	AOAC support part
OPT@	External GPU SKU ID part
UMA@	UMA SKU ID part
14@	For 2410 part
15@	For 2510 part
8106@	8106E LAN part
8111G@	8111G LAN Part
8111GS@	8111GS LAN Part
N14PGV2@	N14P GV2 stuff
GIGA@	GIGA LAN Part
Gastube@	Gastube Part
NOTS@	No Touch screen part
GC6@	GPU GC6 function part
TS@	Touch screen part
@	Not stuff
ME@	Connector
XDP@	XDP part
37@	37W CPU part
47@	47W CPU part
H2@	Hynix 2Gb Vram part
M2@	Micron 2Gb Vram part
M4@	Micron 4Gb Vram part
S2@	Samsung 2Gb Vram part
S4@	Samsung 4Gb Vram part
M1GB@	Micron 1GB Vram BOM
M2GB@	Micron 2GB Vram BOM
S1GB@	Samsung 1GB Vram BOM
S2GB@	Samsung 2GB Vram BOM
H1GB@	hynix 1GB Vram BOM

Fixed Signals				Muxed Signals				Fixed Signals				Muxed Signals				Fixed Signals			
HM86	USB3 1	USB3 2	NA	NA	PCIE 1 (00)	PCIE 2 (00)	PCIE 3	PCIE 4	PCIE 5	PCIE 6	PCIE 7	PCIE 8	SATA 6Gb/s 4 (1b)	SATA 6Gb/s 5 (1b)	SATA 3Gb/s 0	NA	SATA 3Gb/s 2	NA	
					USB3 3 (01)	USB3 4 (01)							PCIE 1 (0b)	PCIE 2 (0b)					
HM87 QM87	USB3 1	USB3 2	USB3 5	USB3 6	PCIE 1 USB3 4	PCIE 2 USB3 5	PCIE 3	PCIE 4	PCIE 5	PCIE 6	PCIE 7	PCIE 8	SATA 6Gb/s 4 PCIE 2	SATA 6Gb/s 5 PCIE 3	SATA 6Gb/s 0	SATA 6Gb/s 1	SATA 3Gb/s 2	SATA 3Gb/s 3	

Soft Strap: (USB3P4_PCIE2_MODE)
00: PCIe Lane 2 is statically assigned to PCIe Express (or GbE)
01: PCIe Lane 2 is statically assigned to USB3 Port 4

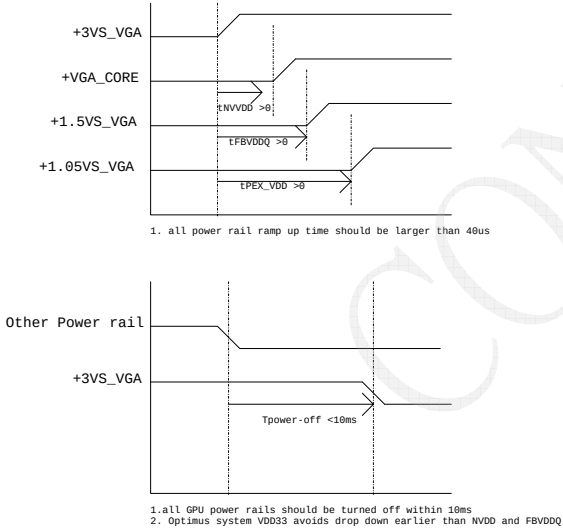
Soft Strap: (USB3P3_PCIE1_MODE)
00: PCIe Lane 1 is statically assigned to PCIe Express (or GbE)
01: PCIe Lane 1 is statically assigned to USB3 Port 3

Config	GPIO16,49
SATA4,SATA5	11
PCIE1,PCIE2	00

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VGA and GDDR3 Voltage Rails (N14P GPIO)

GPIO	I/O	ACTIVE	Function Description
GPIO0	IN	-	FB Clamp monitor
GPIO1	OUT	N/A	
GPIO2	OUT	N/A	
GPIO3	OUT	N/A	
GPIO4	OUT	N/A	
GPIO5		N/A	
GPIO6	OUT	-	Active low FB Clamp toggle request
GPIO7	OUT	N/A	
GPIO8	I/O	-	Thermal Catastrophic Over Temperature
GPIO9	I/O	N/A	2.2K Pull-up
GPIO10	OUT	N/A	
GPIO11	OUT	-	GPU Core VDD PWM control signal
GPIO12	IN		AC Power Detect Input (10K pull High)
GPIO13	OUT	-	Phase Shedding
GPIO14	IN	N/A	
GPIO15	IN	N/A	
GPIO16	OUT	N/A	
GPIO17	IN	N/A	
GPIO18	IN	N/A	
GPIO19	IN	N/A	



Performance Mode P0 TDP at Tj = 102 C* (DDR3)

	GPU (4)	Mem (1.5)	NVCLK /MCLK	NVVDD			FBVDD (1.5V)		FBVDDQ (GPU+Mem) (1.5V)		PCI Express (1.05V) (6)		I/O and PLLVDD (1.8V)		I/O and PLLVDD (1.05V)		Other (3.3V)	
Products	(W)	(W)	(MHz)	(V)	(A)	(W)	(A)	(W)	(A)	(W)	(A)	(W)	(mA)	(W)	(mA)	(W)	(mA)	(W)
N14P 64bit 1GB/2GB DDR3	25W	TBD	1000MHz	TBD	32	TBD	1.7	2.55	TBD	TBD	1.98	2.1	TBD	TBD	TBD	TBD	TBD	TBD

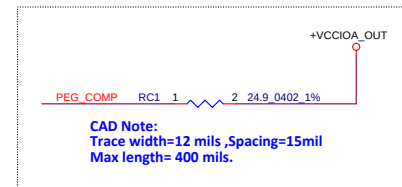
Physical Strapping pin	Power Rail	Logical Strapping Bit3	Logical Strapping Bit2	Logical Strapping Bit1	Logical Strapping Bit0
ROM_SCLK	+3VS_VGA	PCI_DEVID[4]	SUB_VENDOR	PCI_DEVID5]	PEX_PLL_EN_TERM
ROM_SI	+3VS_VGA	RAM_CFG[3]	RAM_CFG[2]	RAM_CFG[1]	RAM_CFG[0]
ROM_S0	+3VS_VGA	FB[1]	FB[0]	SMB_ALT_ADDR	VGA_DEVICE
STRAP0	+3VS_VGA	USER[3]	USER[2]	USER[1]	USER[0]
STRAP1	+3VS_VGA	3GIO_PAD_CFG_ADR[3]	3GIO_PAD_CFG_ADR[2]	3GIO_PAD_CFG_ADR[1]	3GIO_PAD_CFG_ADR[0]
STRAP2	+3VS_VGA	PCI_DEVID[3]	PCI_DEVID[2]	PCI_DEVID[1]	PCI_DEVID[0]
STRAP3	+3VS_VGA	SOR3_EXPOSED	SOR2_EXPOSED	SOR1_EXPOSED	SOR0_EXPOSED
STRAP4	+3VS_VGA	RESERVED	PCIE_SPEED_CHANGE_GEN3	PCIE_MAX_SPEED	DP_PLL_VDD33V

GPU	Device ID	SMB_ALT_ADDR (ROM_S0 Bit 1)	setting	I2C Slave addrees ID
N14P-GV2	0x1292		0	0x9E (Default)
			1	0x9C

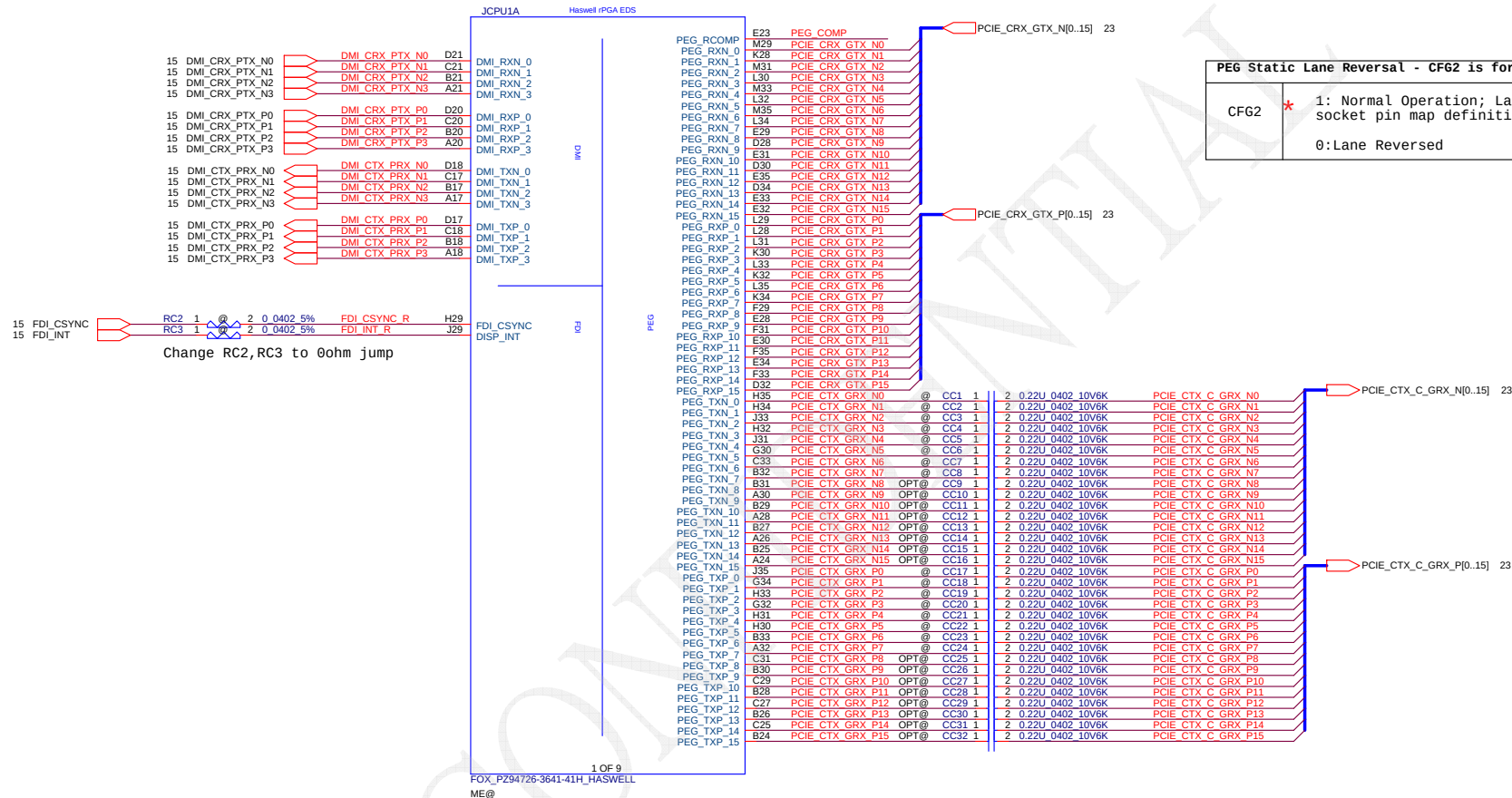
GPU	ROM_SI	ROM_S0	ROM_SCLK	STRAP0	STRAP1	STRAP2	STRAP3	STRAP4
N14P-GV2	TBD	PU 4.99K	PU 4.99K	PU 45.3K	PD 45.3K	PD 15K	PD 4.99K	PU 45.3K

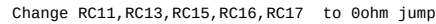
GPU		N14P_GV2		
FB Memory (DDR3)		ROM_SI		
Samsung 1GHz	K4W2G1646E-BC1A	0x7		
	128M x 16	PD 45.3K		
Micron 1GMHz	MT41J128M16JT-0936:K	0x5		
	128M x 16	PD 30.1K		
Hynix 1GMHz	H5TC2663FFR-11C	0x4		
	128M x 16	PD 24.9K		
Samsung 900MHz	K4W4G1646B-HC11	0x3		
	256M x 16	PD 20K		
Micron 900MHz	MT41K256M16HA-1076:E	0x1		
	256M x 16	PD 10K		

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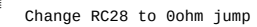
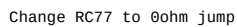
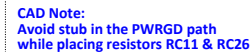
PEG Static Lane Reversal - CF62 is for the 16x	
CF62	* 1: Normal Operation; Lane # definition matches socket pin map definition 0: Lane Reversed



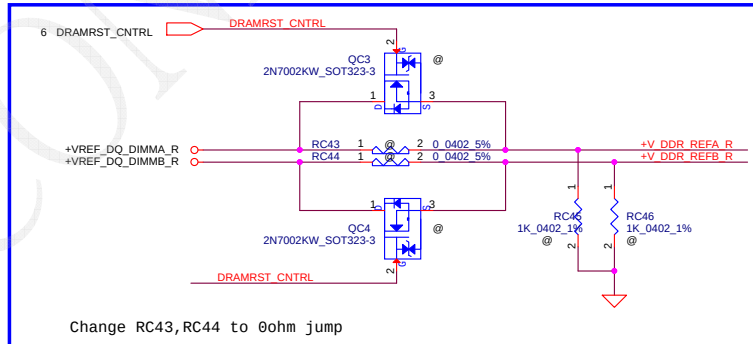
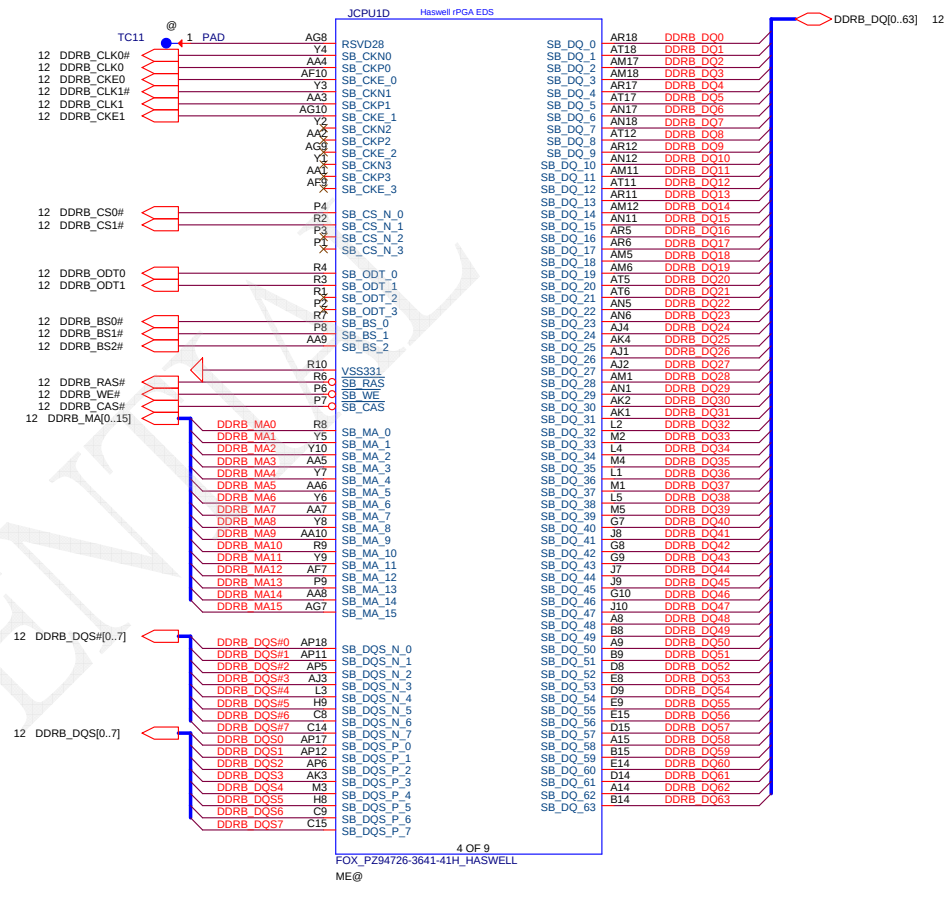
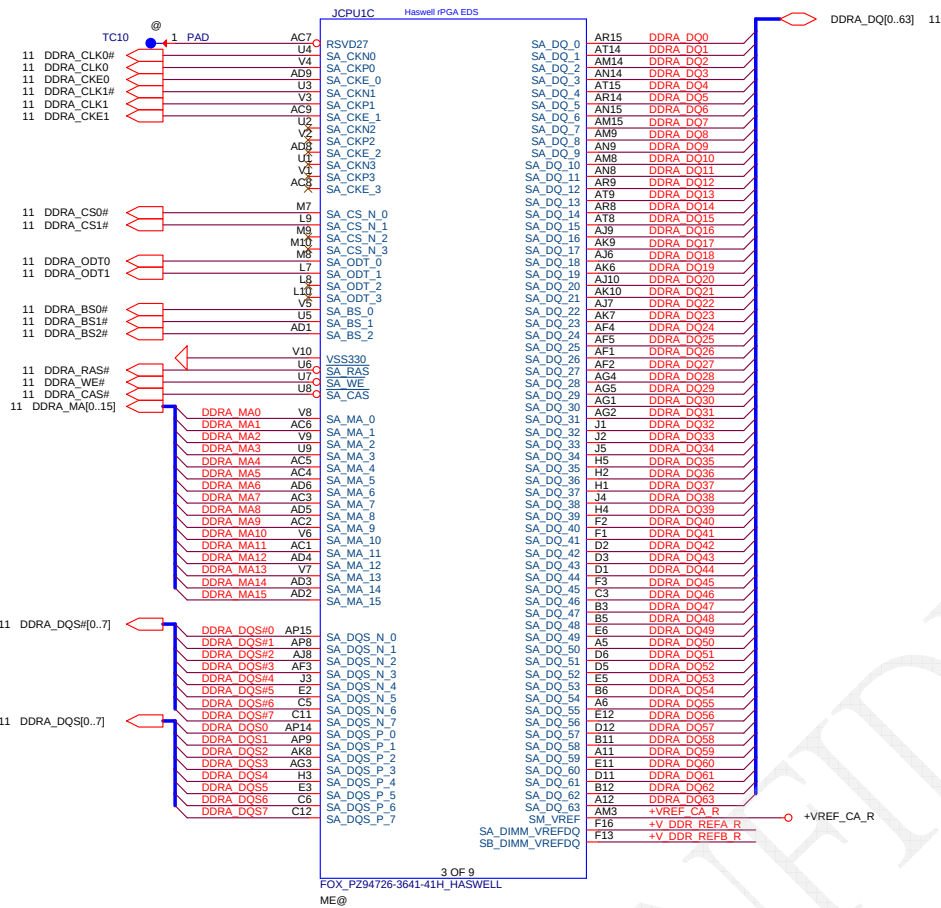


19 CPU_PLTRST#

For ESD concern, please put near CPU



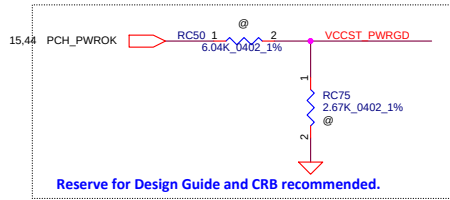
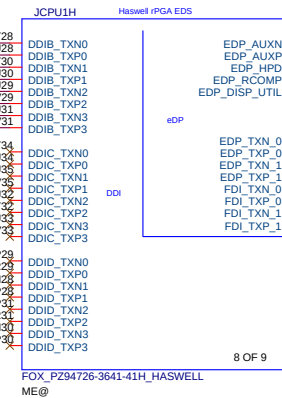
Title			
CPU (2/7) PM, XDP, CLK			
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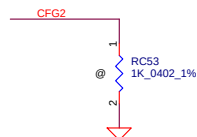
HDMI D2
HDMI D1
HDMI D0
HDMI CLK

35 HDMI_TX2+
35 HDMI_TX2-
35 HDMI_TX1+
35 HDMI_TX1-
35 HDMI_TX0+
35 HDMI_TX0-
35 HDMI_CLK+
35 HDMI_CLK-



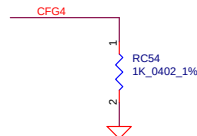
Reserve for Design Guide and CRB recommended.

CFG STRAPS for CPU



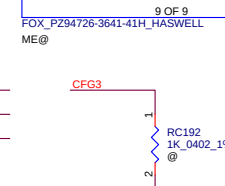
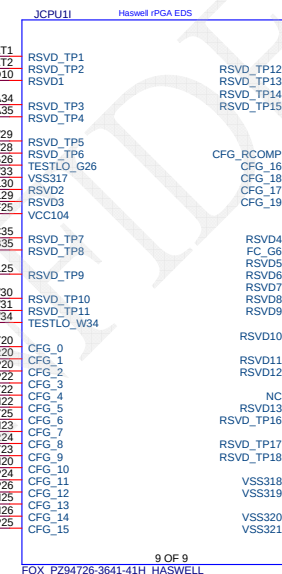
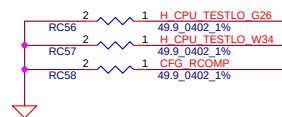
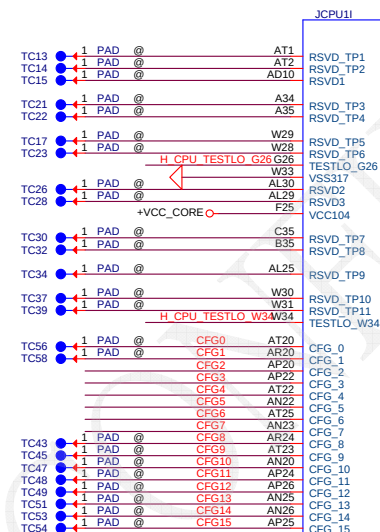
PEG Static Lane Reversal - CFG2 is for the 16x

CFG2	1: (Default) Normal Operation; Lane # definition matches socket pin map definition 0: Lane Reversed
------	--



Display Port Presence Strap

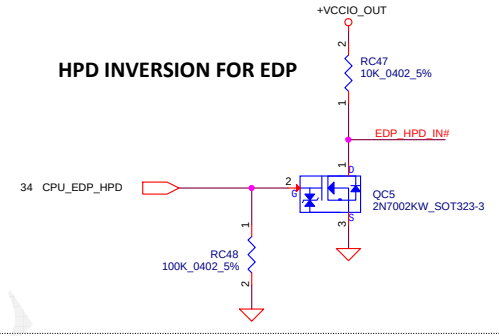
CFG4	1 : Disabled; No Physical Display Port attached to Embedded Display Port * 0 : Enabled; An external Display Port device is connected to the Embedded Display Port
------	--



PHYSICAL_DEBUG_ENABLED (DFX PRIVACY)

CFG3	0 : ENABLED SET DFX_ENABLED BIT IN DEBUG INTERFACE MSR * 1 : DISABLED
------	---

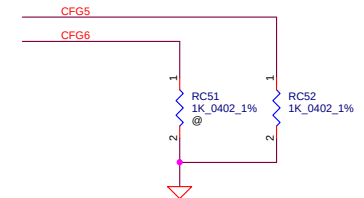
HPD INVERSION FOR EDP



COMPENSATION PU FOR eDP

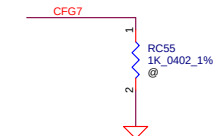


CAD Note: Trace width=20 mils, Spacing=25mil, Max length=100 mils.



PCIe Port Bifurcation Straps

CFG[6:5]	11: (Default) x16 - Device 1 functions 1 and 2 disabled * 10: x8, x8 - Device 1 function 1 enabled; function 2 disabled 01: Reserved - (Device 1 function 1 disabled; function 2 enabled) 00: x8, x4, x4 - Device 1 functions 1 and 2 enabled
----------	--



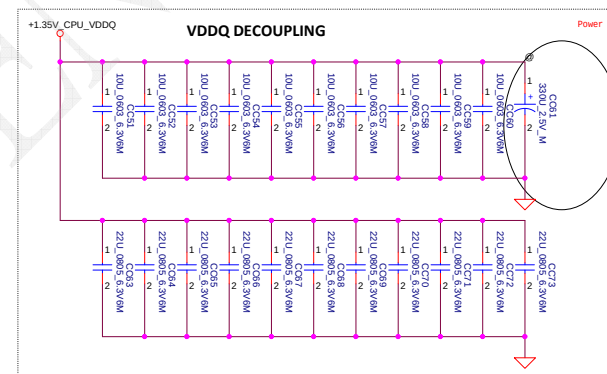
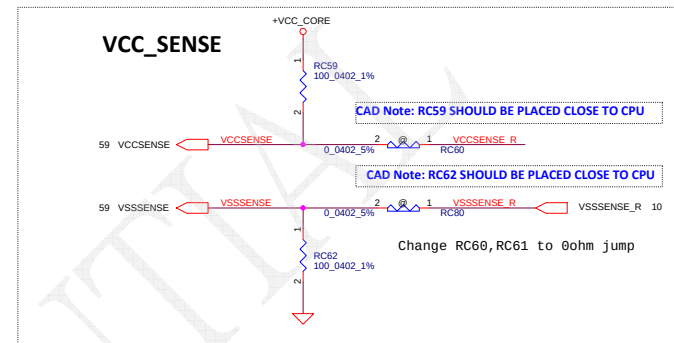
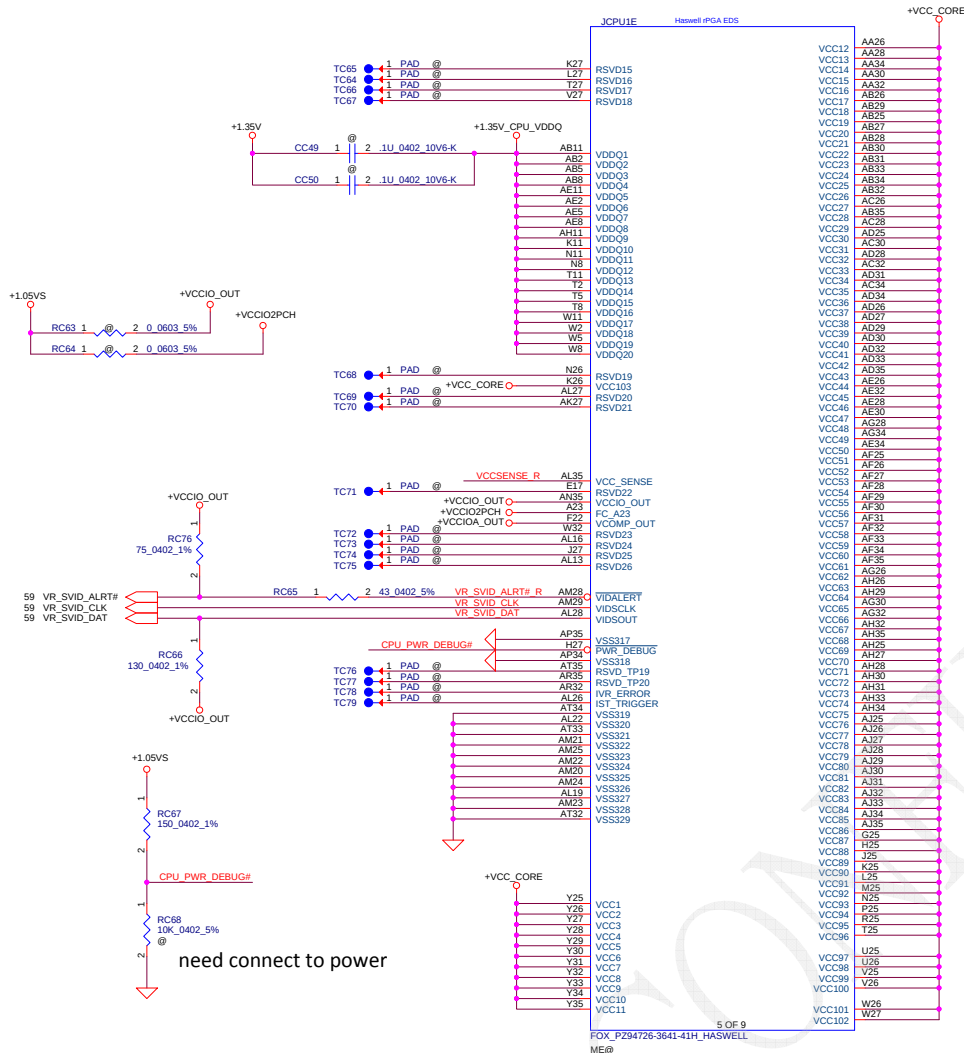
PEG DEFER TRAINING

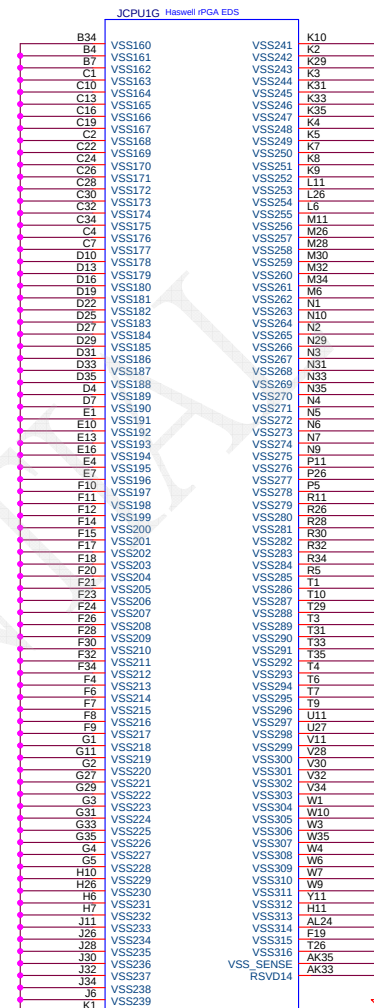
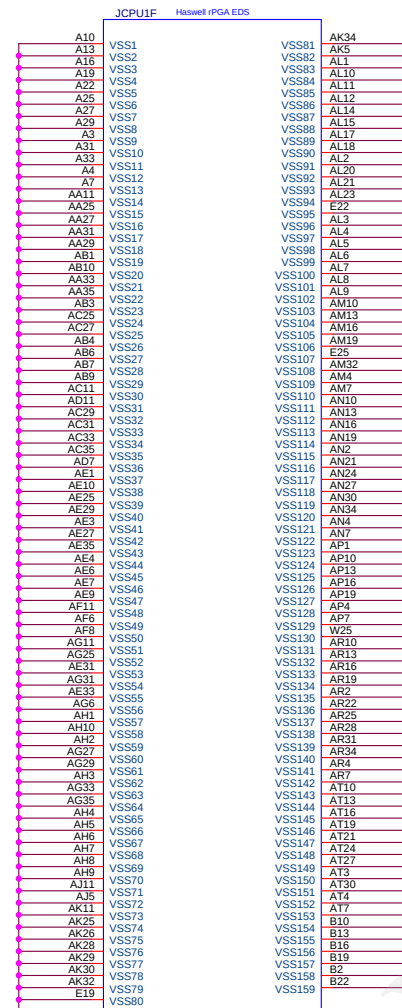
CFG7	* 1: (Default) PEG Train immediately following xxRESET de assertion 0: PEG Wait for BIOS for training
------	--

Need confirm with Intel if this reserved circuit can be deleted.

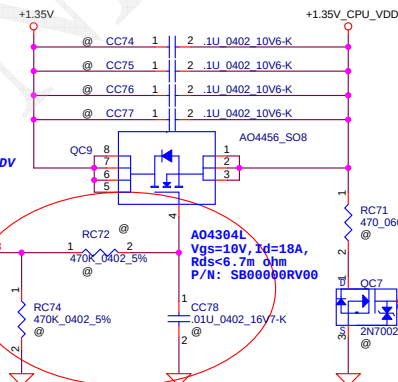
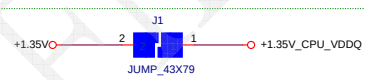
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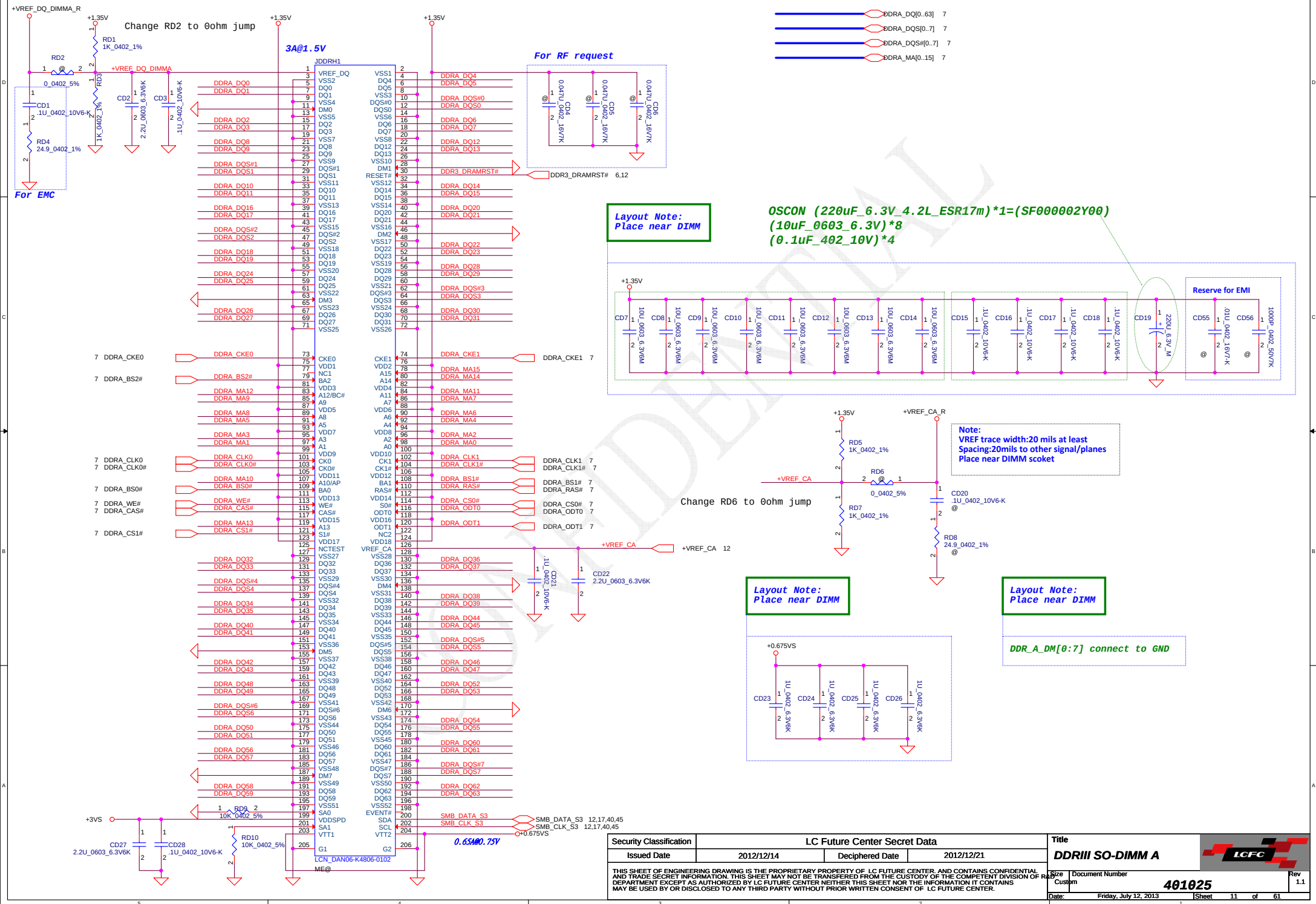
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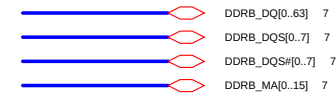
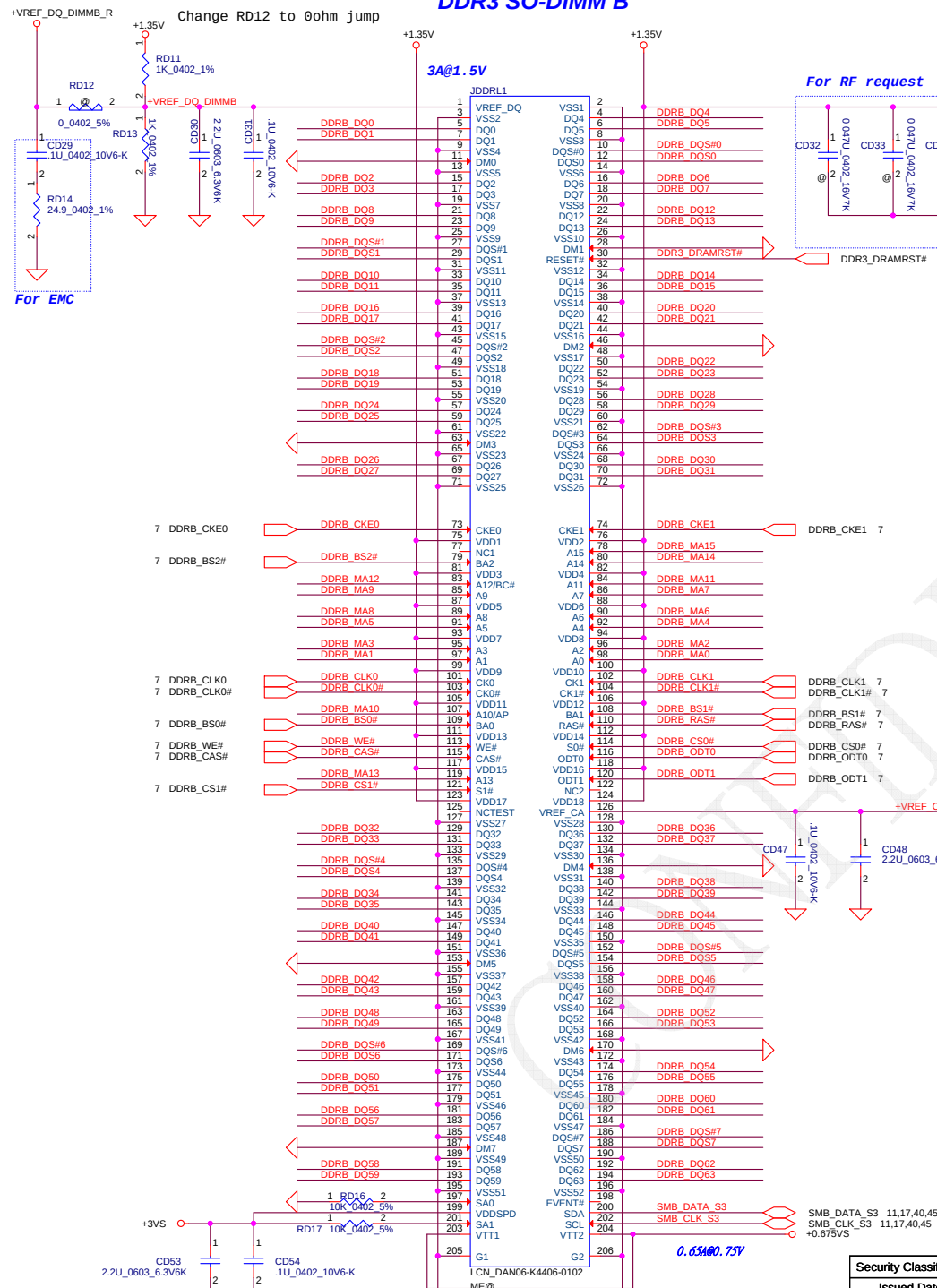




+1.35V_CPU_VDDQ

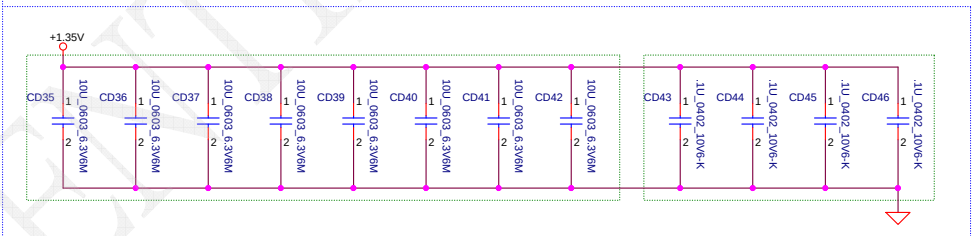


DDR3 SO-DIMM A

DDR3 SO-DIMM B

Layout Note:
Place near DIMM

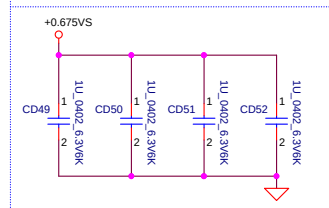
(10uF_0603_6.3V)*8
(0.1uF_402_10V)*4



Layout Note:
Place near DIMM

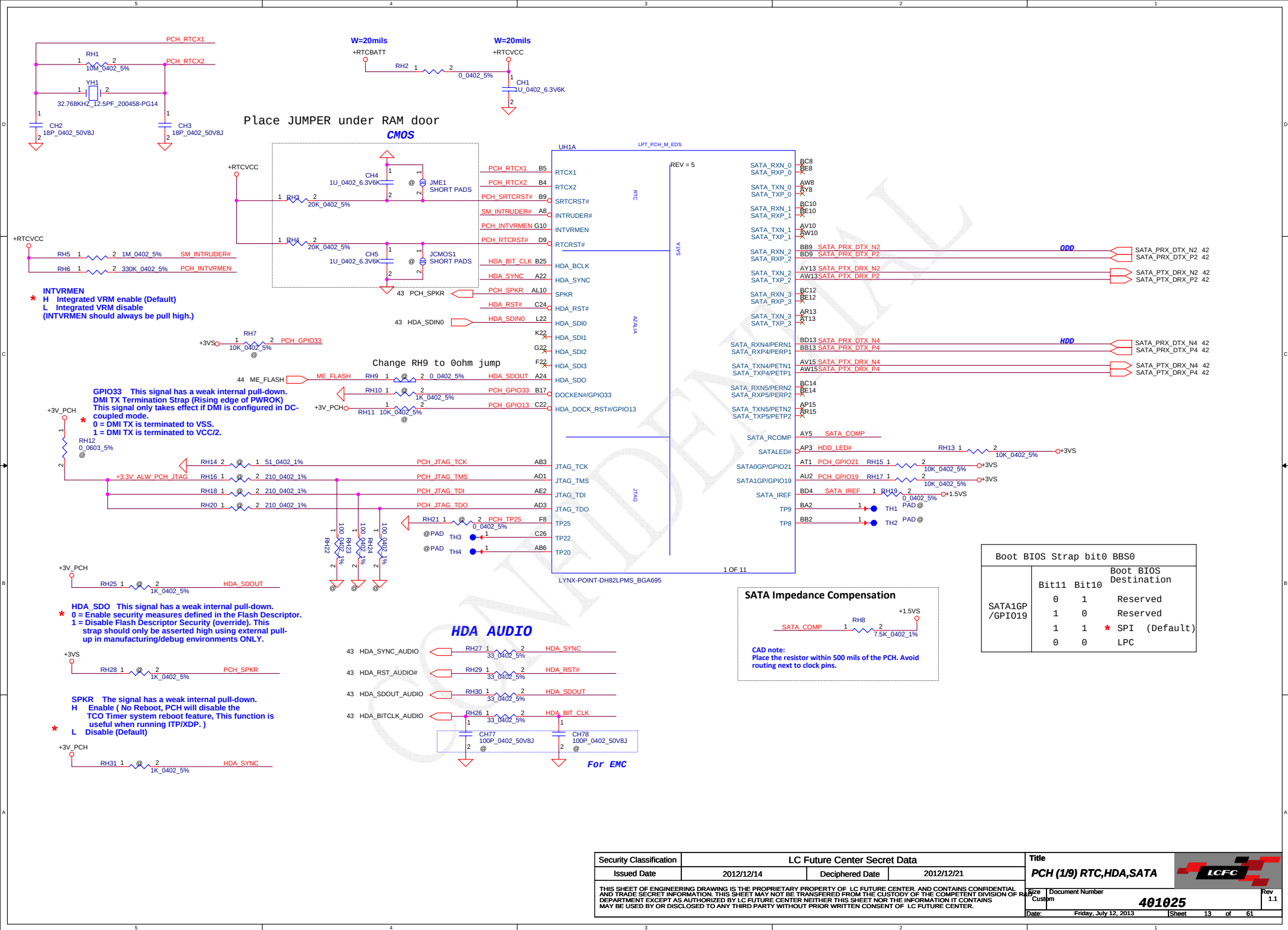
Layout Note:
Place near DIMM

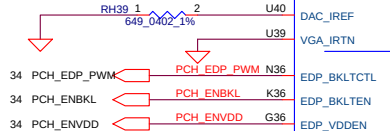
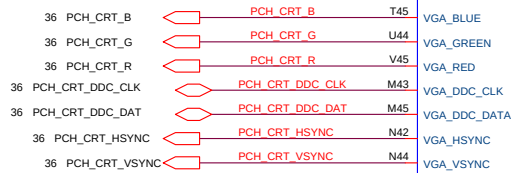
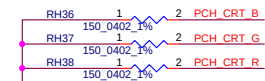
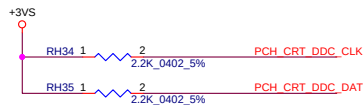
DDR_B_DM[0:7] connect to GND



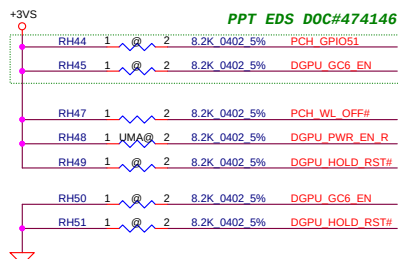
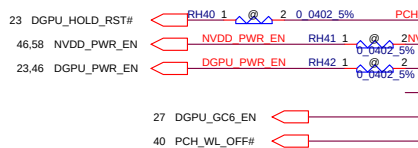
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Title			
DDRIII SO-DIMM B			
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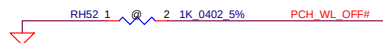




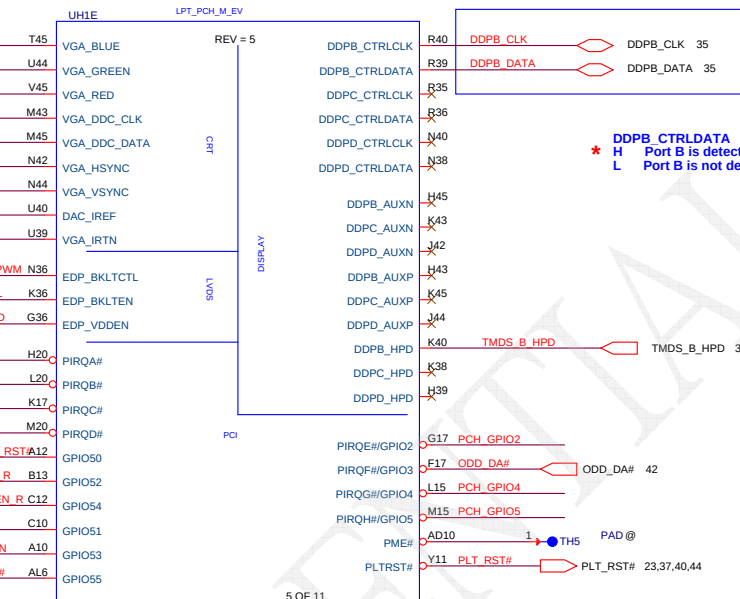
Change RH40, RH41, RH42 to 0ohm jump



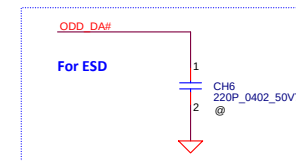
GPIO53 The signal has a weak internal pull-up.
H DMI is in DC-coupling mode (desktop, mobile or server/workstation).
L DMI is in AC-coupling mode (server/workstation only, not meant for desktop/mobile).



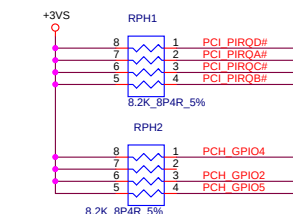
GPIO55 The signal has a weak internal pull-up.
H Disable 'Top-Block Swap' mode.
L Enable 'Top-Block Swap' mode.

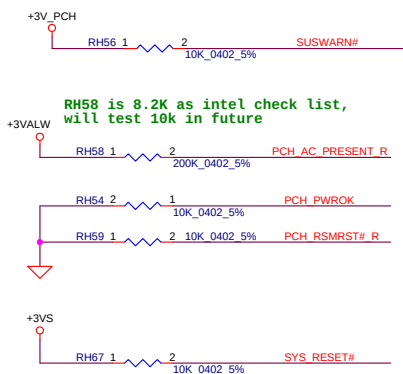
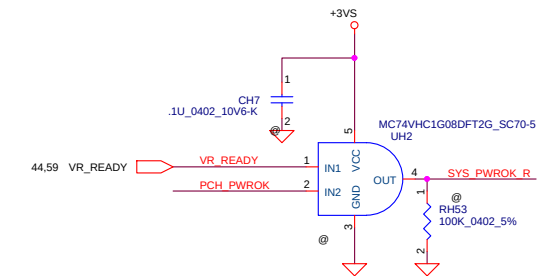


* DDPB_CTRLCLK The signal has a weak internal pull-down.
H Port B is detected.
L Port B is not detected.

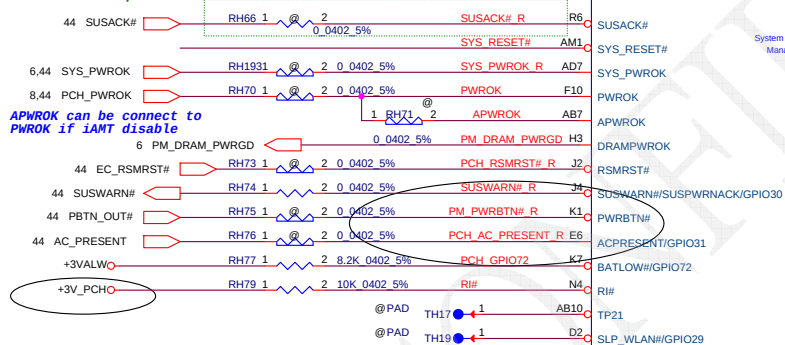


Boot BIOS Strap		
BBS_BIT1 (GPIO51)	SATA_SLPD (BBS_BIT0)	Boot BIOS Location
0	0	LPC
0	1	Reserved (NAND)
1	0	PCI
1	1	SPI



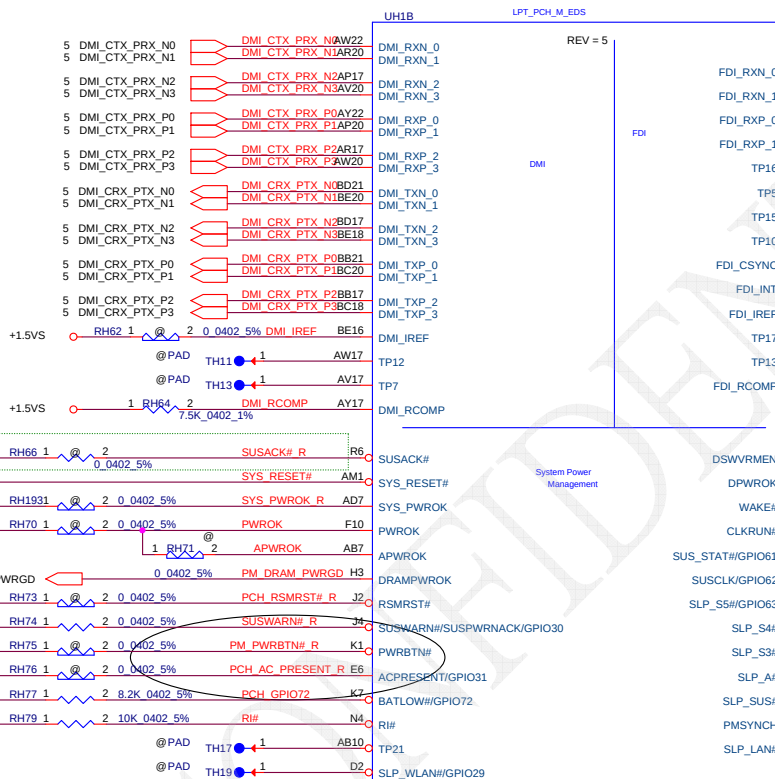
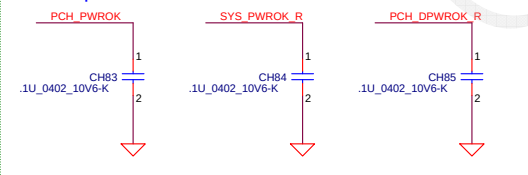


For Deep S3

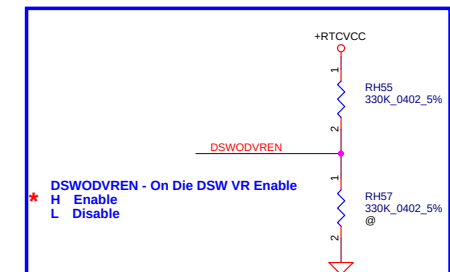
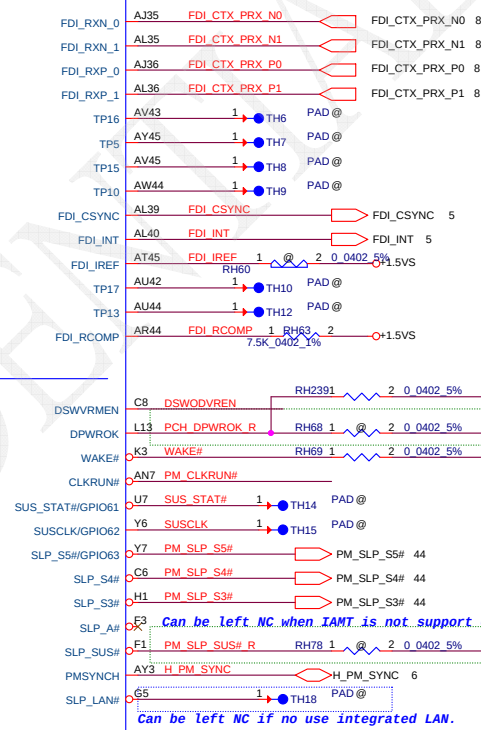


Change RH62, RH193, RH70, RH71, RH73, RH75, RH76, RH60 to 0ohm jump

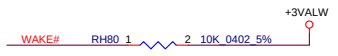
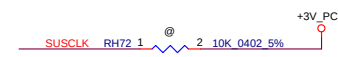
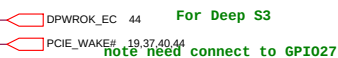
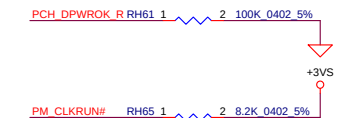
AS EMC request



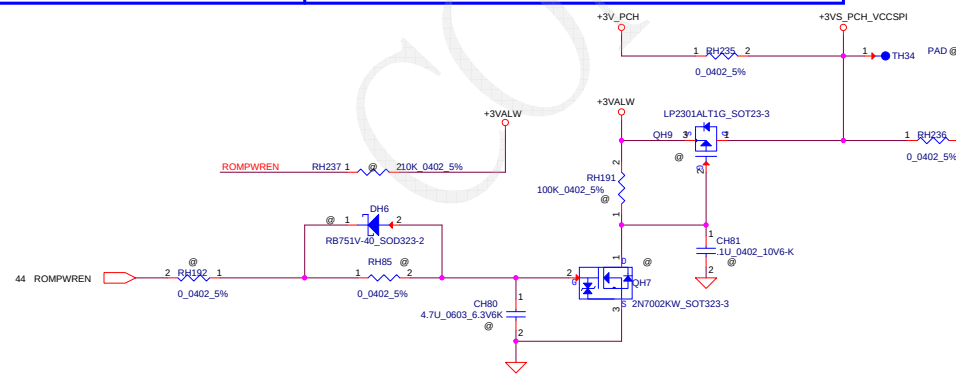
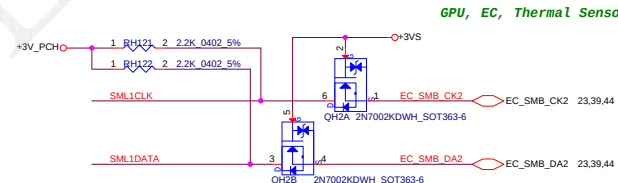
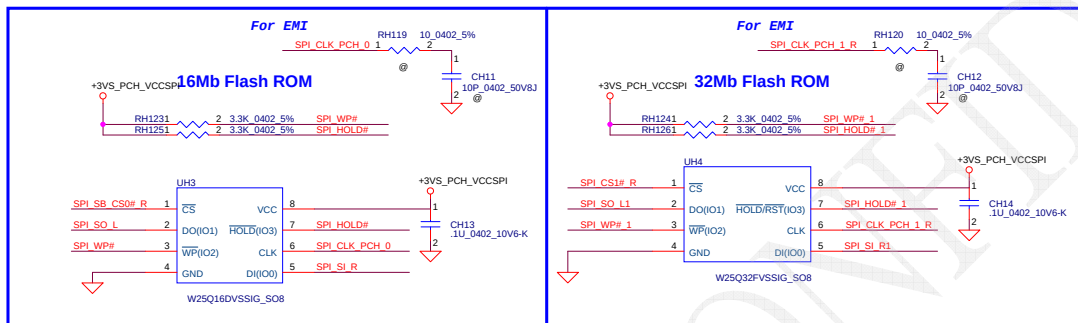
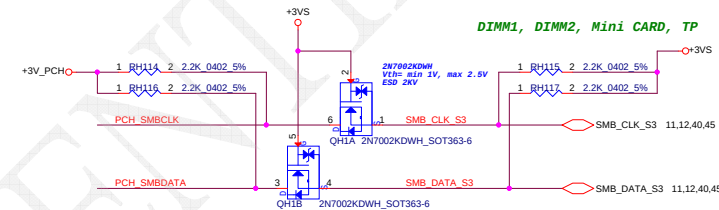
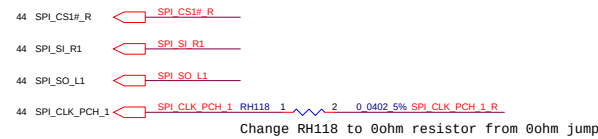
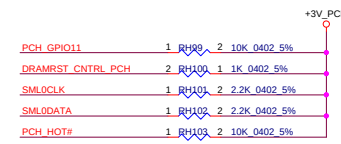
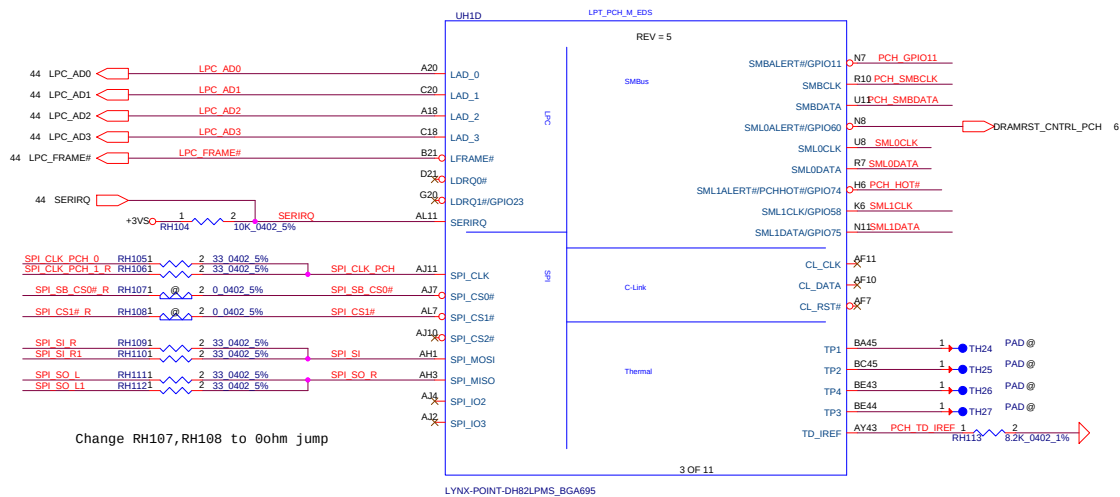
LYNX-POINT-DH82LPMS_BGA695

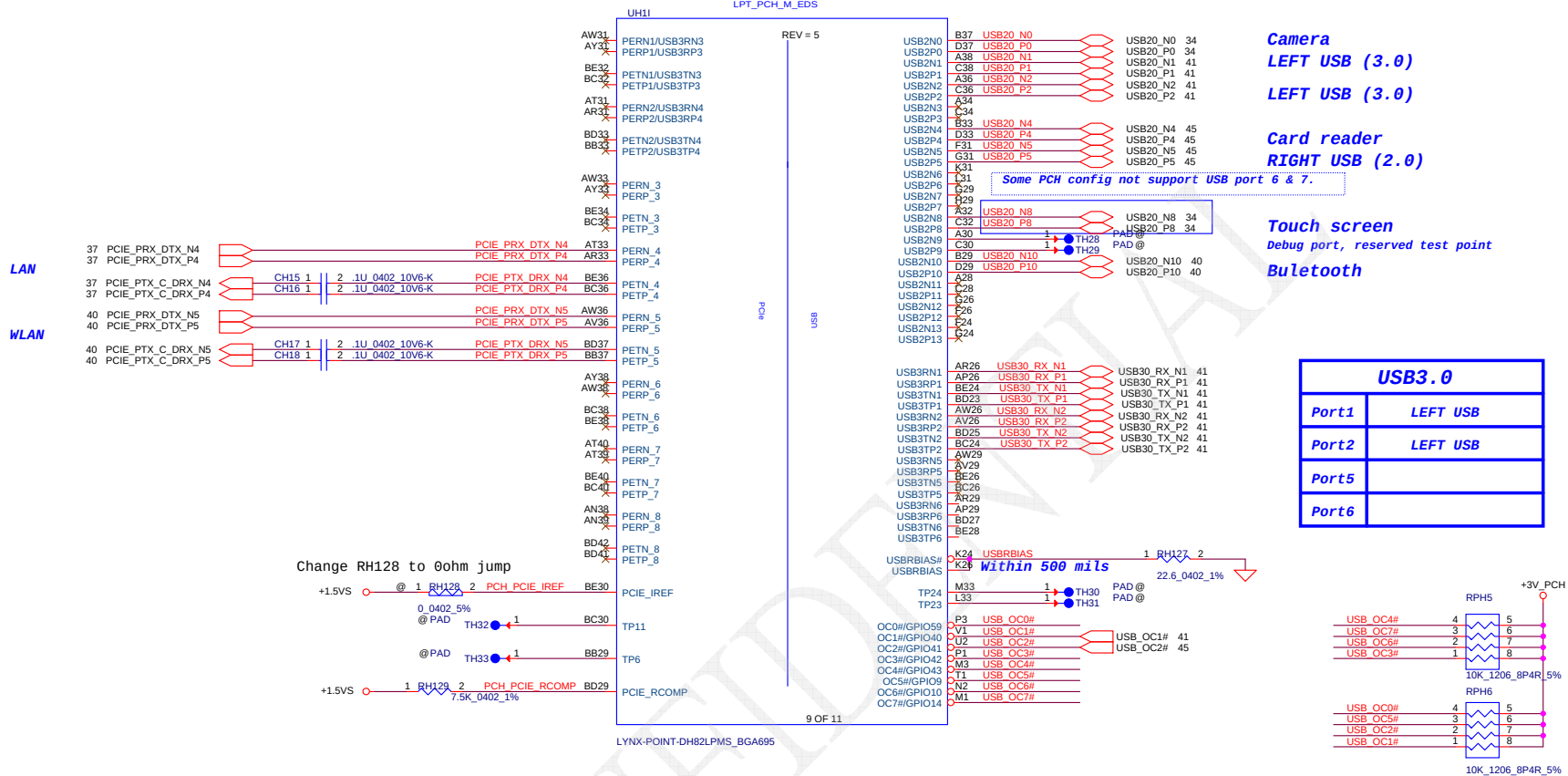


RH61 is 1% as check list request, CRB is 5%. follow CRB



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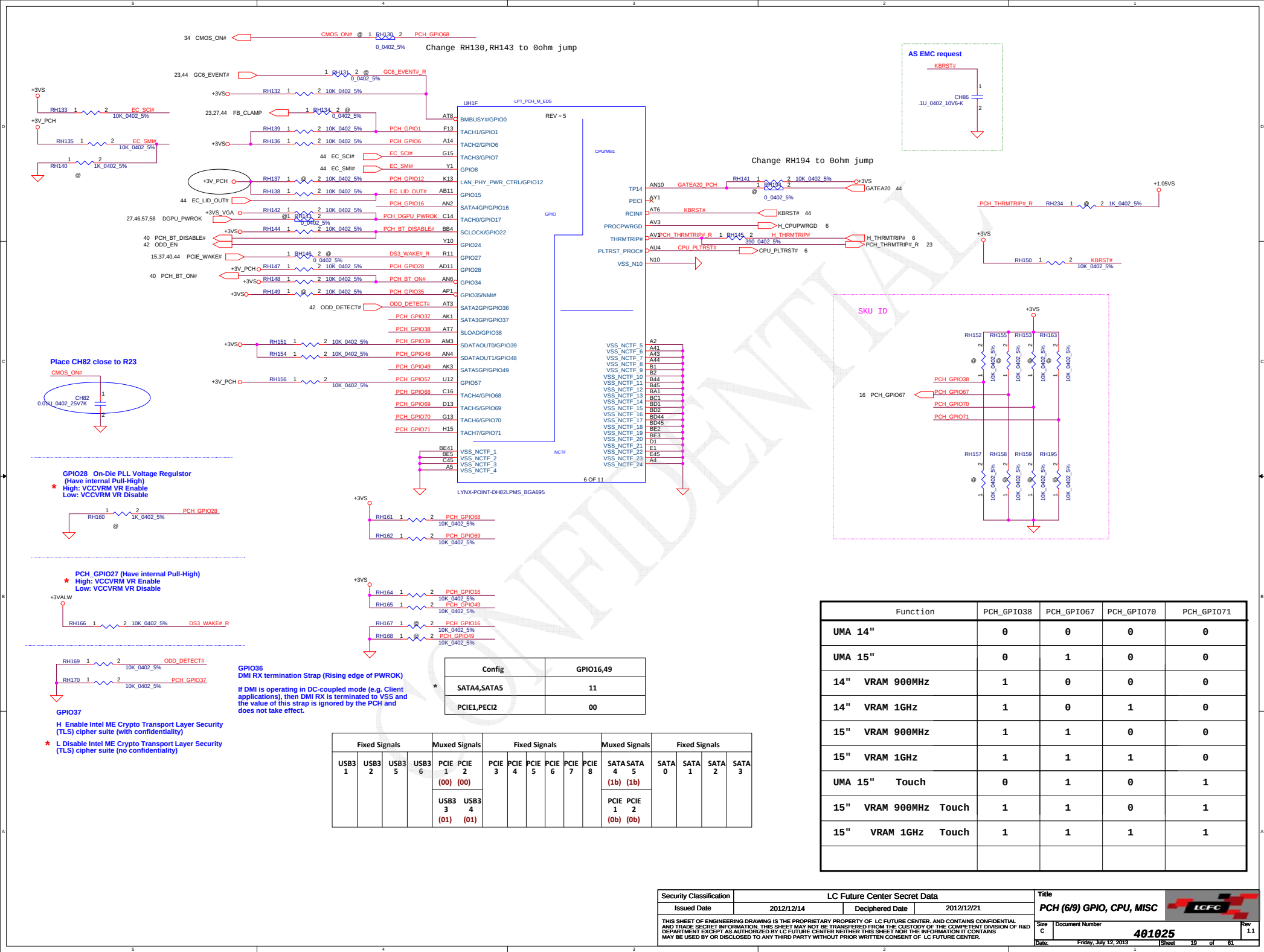


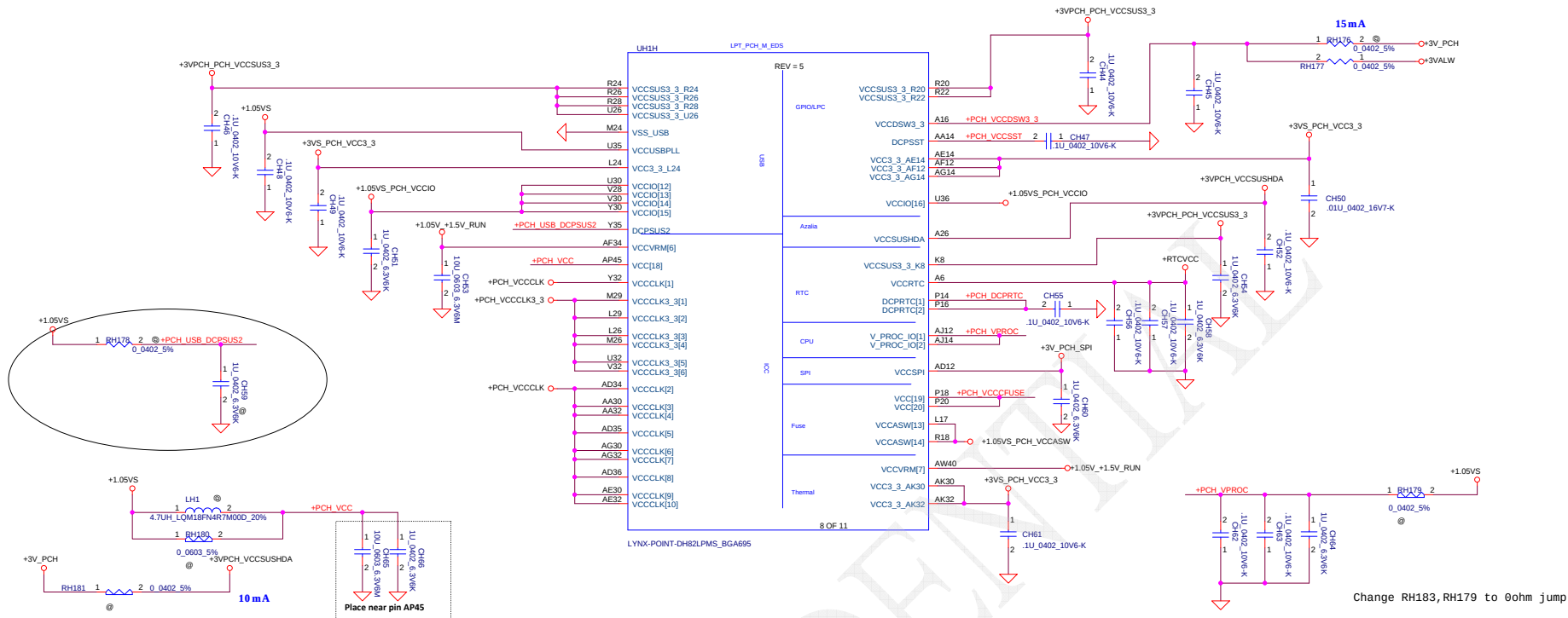


Camera
LEFT USB (3.0)
LEFT USB (3.0)

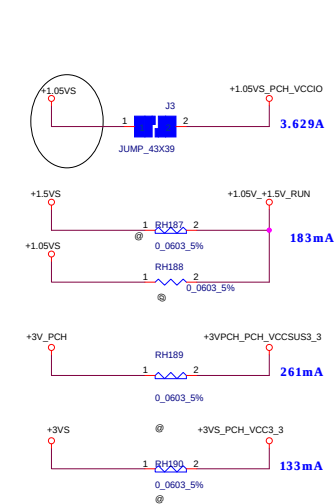
Card reader
RIGHT USB (2.0)

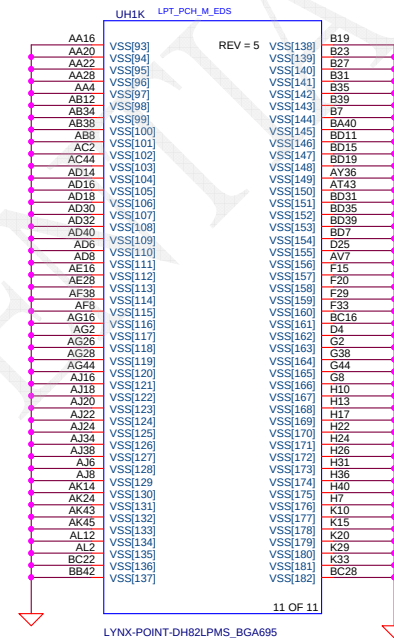
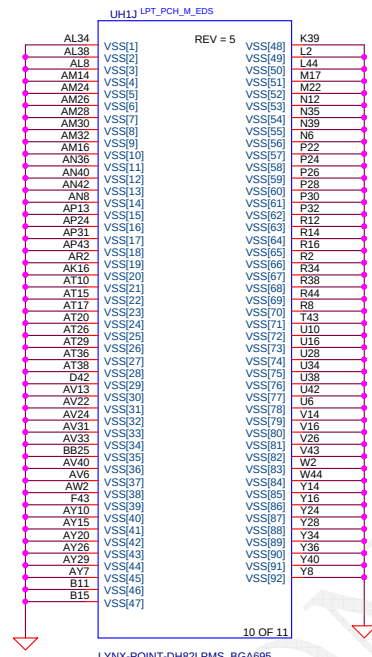
Touch screen
Debug port, reserved test point
Buletooth

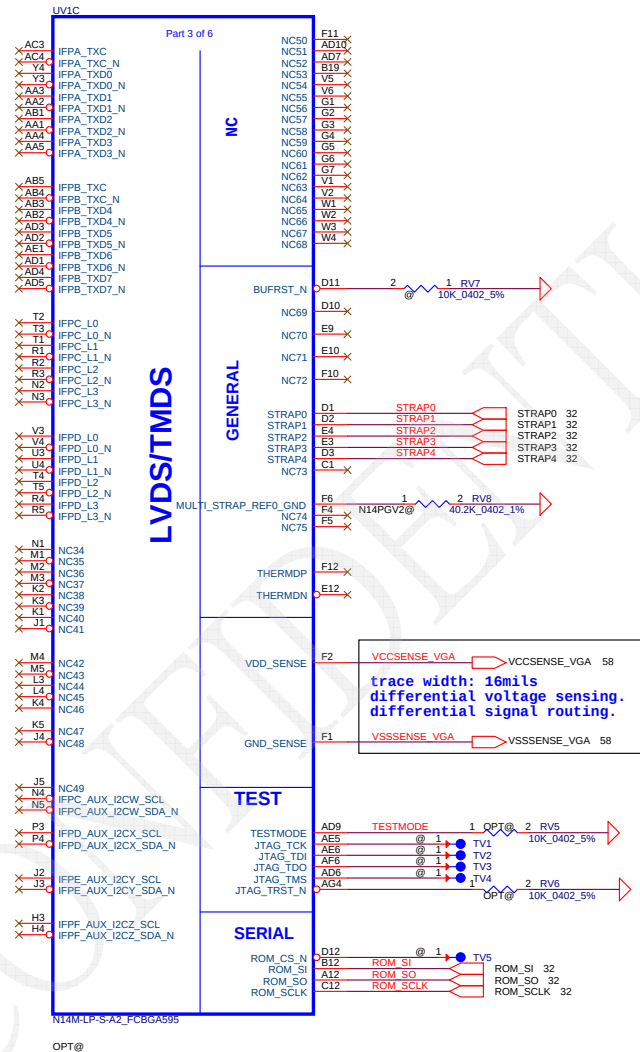


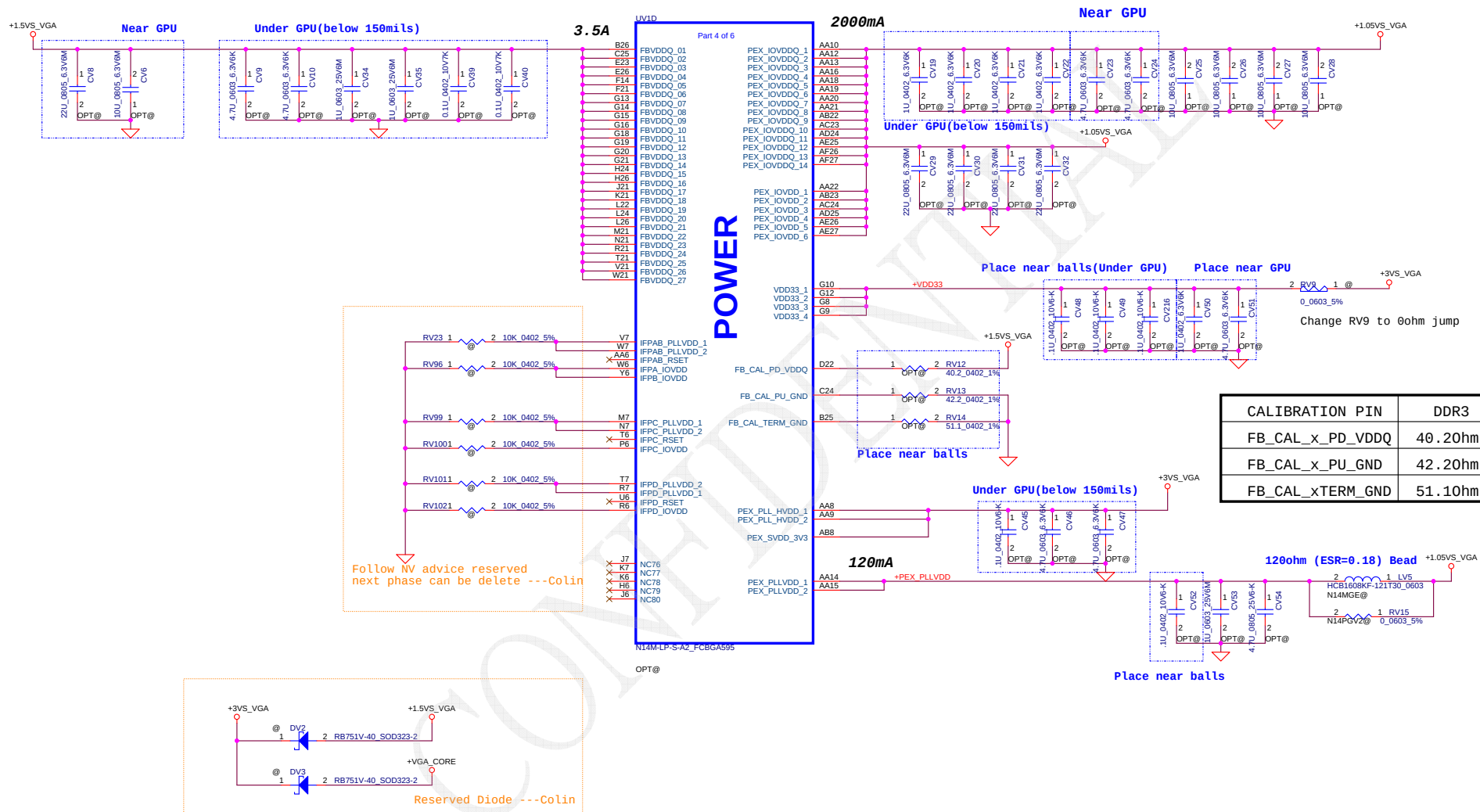


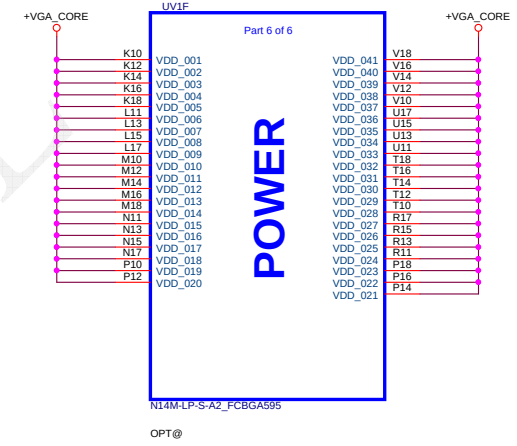
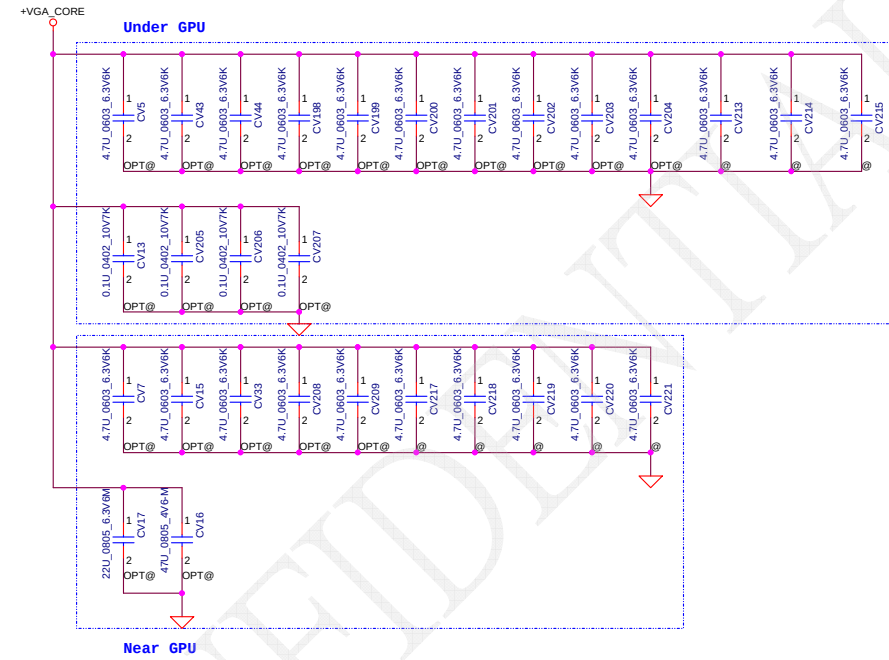
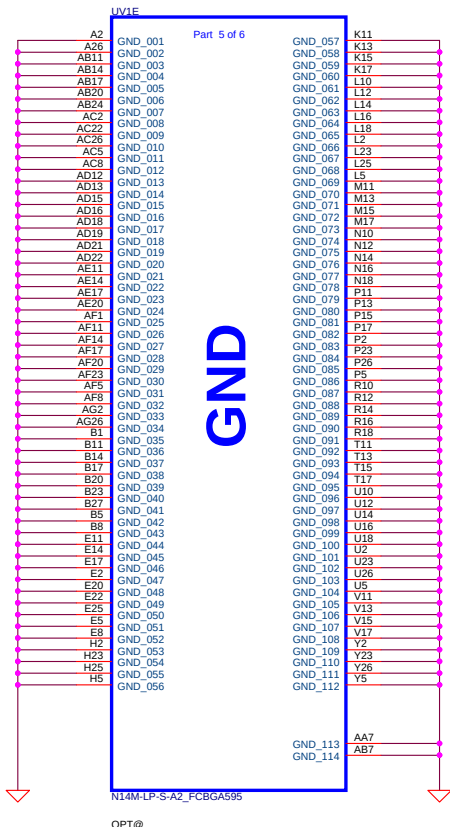
Change RH180, RH181, RH187, RH189, RH190, RH185, RH186 to 0ohm jump





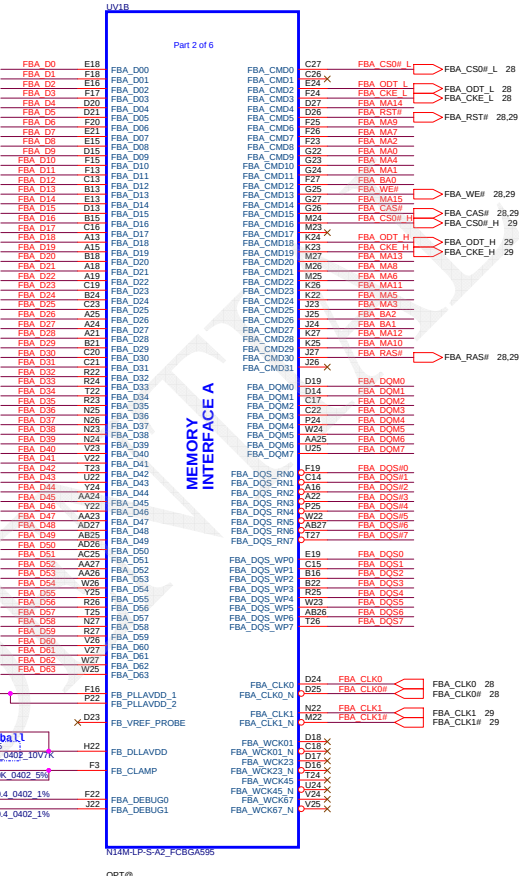
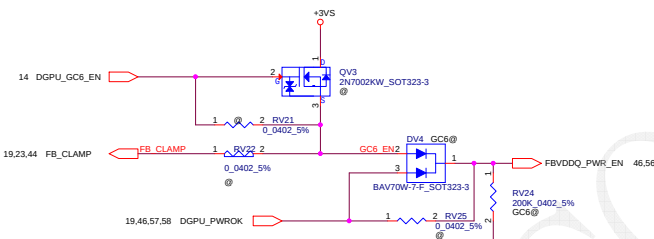
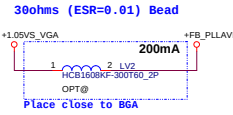
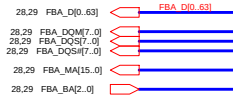




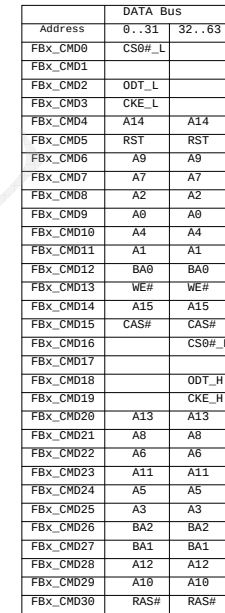
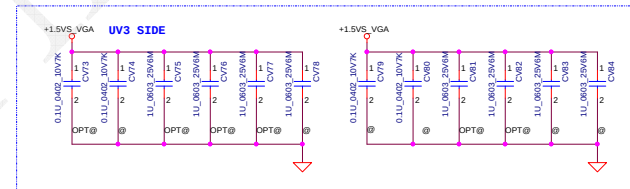
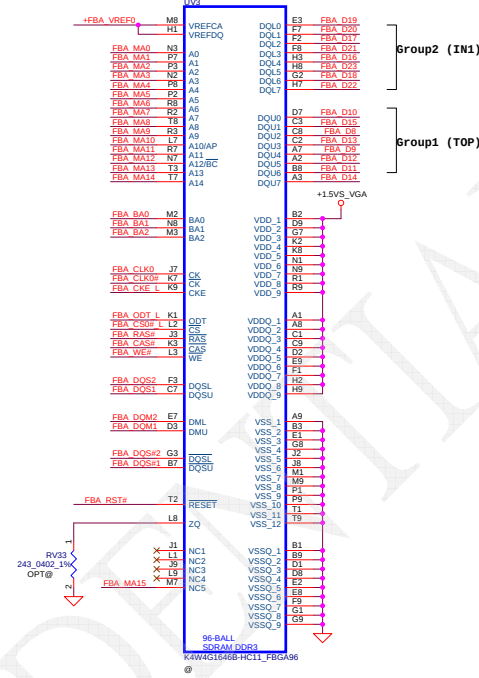
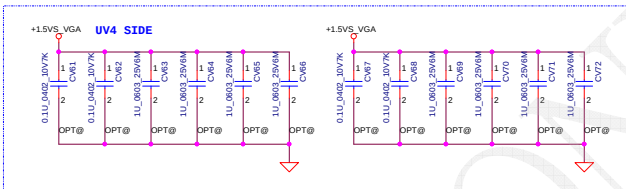
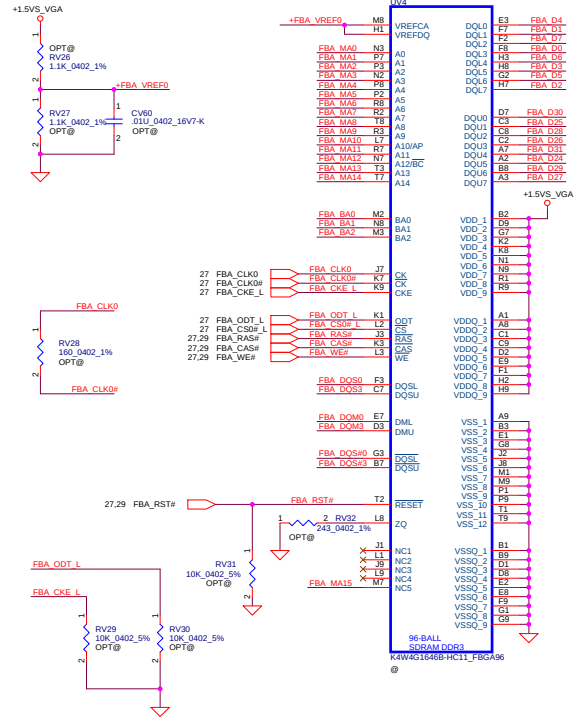


Mode D - Mirror Mode Mapping

	DATA Bus	
Address	0..31	32..63
FBx_CMD0	CS0#_L	
FBx_CMD1		
FBx_CMD2	ODT_L	
FBx_CMD3	CKE_L	
FBx_CMD4	A14	A14
FBx_CMD5	RST	RST
FBx_CMD6	A9	A9
FBx_CMD7	A7	A7
FBx_CMD8	A2	A2
FBx_CMD9	A0	A0
FBx_CMD10	A4	A4
FBx_CMD11	A1	A1
FBx_CMD12	BA0	BA0
FBx_CMD13	WE#	WE#
FBx_CMD14	A15	A15
FBx_CMD15	CAS#	CAS#
FBx_CMD16	CS0#_H	
FBx_CMD17		
FBx_CMD18	ODT_H	
FBx_CMD19	CKE_H	
FBx_CMD20	A13	A13
FBx_CMD21	A8	A8
FBx_CMD22	A6	A6
FBx_CMD23	A11	A11
FBx_CMD24	A5	A5
FBx_CMD25	A3	A3
FBx_CMD26	BA2	BA2
FBx_CMD27	BA1	BA1
FBx_CMD28	A12	A12
FBx_CMD29	A10	A10
FBx_CMD30	RAS#	RAS#

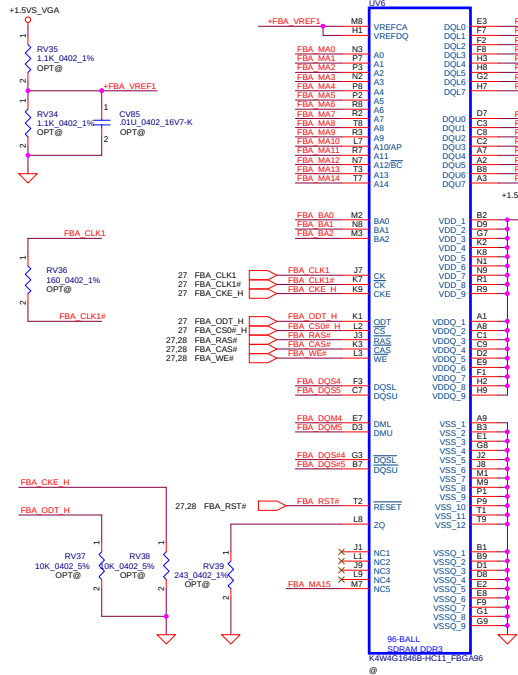


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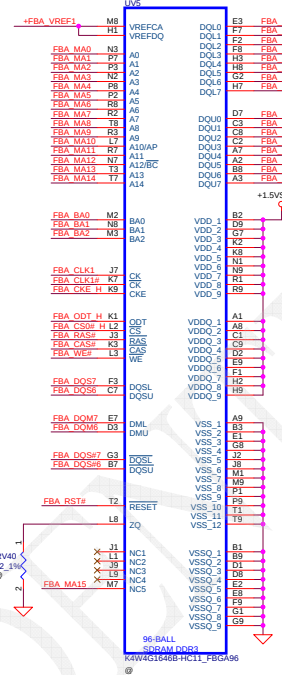
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at least 16 mils width(optimal)
20 mils spacing to other signals /planes



Group4 (IN1)

Group5 (TOP)

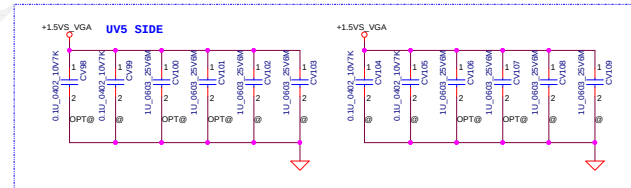
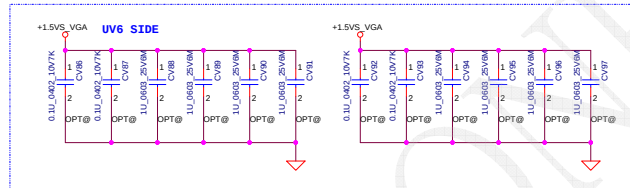


Group7 (IN3)

Group6 (BOT)

CMD mapping mod Mode D

Address	DATA Bus
0...31	32...63
FBx_CMD0	CS0#_L
FBx_CMD1	
FBx_CMD2	ODT_L
FBx_CMD3	CKE_L
FBx_CMD4	A14
FBx_CMD5	RST
FBx_CMD6	A9
FBx_CMD7	A7
FBx_CMD8	A2
FBx_CMD9	A0
FBx_CMD10	A4
FBx_CMD11	A1
FBx_CMD12	BA0
FBx_CMD13	WE#
FBx_CMD14	A15
FBx_CMD15	CAS#
FBx_CMD16	CS0#_H
FBx_CMD17	
FBx_CMD18	ODT_H
FBx_CMD19	CKE_H
FBx_CMD20	A13
FBx_CMD21	A8
FBx_CMD22	A6
FBx_CMD23	A11
FBx_CMD24	A5
FBx_CMD25	A3
FBx_CMD26	BA2
FBx_CMD27	BA1
FBx_CMD28	A12
FBx_CMD29	A10
FBx_CMD30	RAS#



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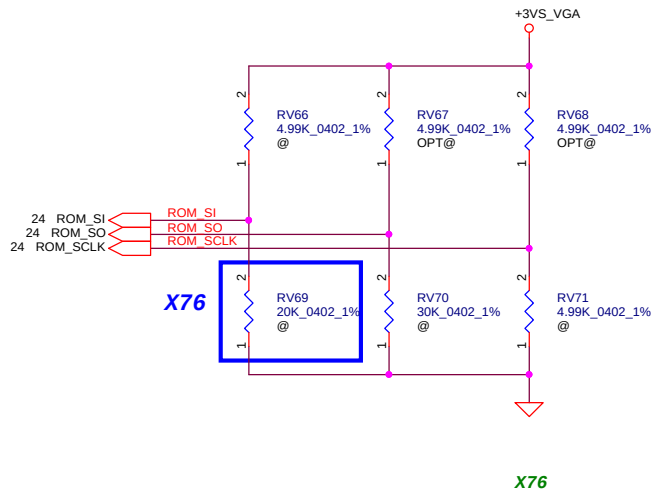
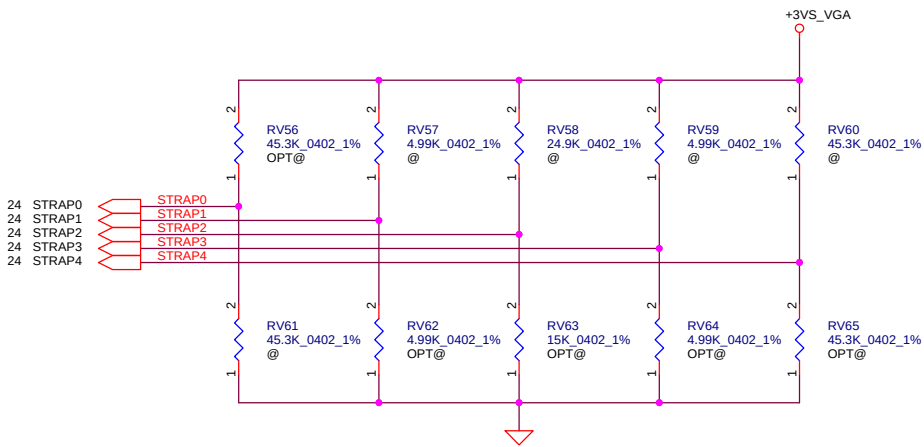
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X76

Physical Strapping pin	Power Rail	Logical Strapping Bit3	Logical Strapping Bit2	Logical Strapping Bit1	Logical Strapping Bit0
ROM_SCLK	+3VS_VGA	PCI_DEVID[4]	SUB_VENDOR	PCI_DEVID5]	PEX_PLL_EN_TERM
ROM_SI	+3VS_VGA	RAM_CFG[3]	RAM_CFG[2]	RAM_CFG[1]	RAM_CFG[0]
ROM_SO	+3VS_VGA	FB[1]	FB[0]	SMB_ALT_ADDR	VGA_DEVICE
STRAP0	+3VS_VGA	USER[3]	USER[2]	USER[1]	USER[0]
STRAP1	+3VS_VGA	3GIO_PAD_CFG_ADR[3]	3GIO_PAD_CFG_ADR[2]	3GIO_PAD_CFG_ADR[1]	3GIO_PAD_CFG_ADR[0]
STRAP2	+3VS_VGA	PCI_DEVID[3]	PCI_DEVID[2]	PCI_DEVID[1]	PCI_DEVID[0]
STRAP3	+3VS_VGA	SOR3_EXPOSED	SOR2_EXPOSED	SOR1_EXPOSED	SOR0_EXPOSED
STRAP4	+3VS_VGA	RESERVED	PCIE_SPEED_CHANGE_GEN3	PCIE_MAX_SPEED	DP_PLL_VDD33V

Resistor Values	Pull-up to +3VS_VGA	Pull-down to Gnd
4.99K	1000	0000
10K	1001	0001
15K	1010	0010
20K	1011	0011
24.9K	1100	0100
30.1K	1101	0101
34.8K	1110	0110
45.3K	1111	0111

FB[1:0]	
0	Reserved
1	Reserved
2	256MB (Default)
3	Reserved

PCIE_MAX_SPEED	
0	Limit booting to PCIe Gen1
1	Allow booting to PCIe Gen 2/3

VGA_DEVICE	
0	3D Device (Class Code 302h)
1	VGA Device (Default)

SMBUS_ALT_ADDR	
0	0x9E (Default)
1	0x9C (Multi-GPU usage)

SUB_VENDOR	
0	No VBIOS ROM
1	BIOS ROM is present (Default)

PEX_PLL_EN_TERM	
0	Disable (Default)
1	Enable

PCIE_SPEED_CHANGE_GEN3	
0	Disable PCIe Gen3 operation
1	Enable PCIe Gen3 operation

USER Straps	
User[3:0]	
1000-1100	Customer defined

DP_PLL_VDD33V	
0	Reserved
1	Default

3GIO_PADCFG[3:0]	
0110	Gen1/Gen2 support only
0000	Gen3 support

GPU	FB Memory (DDR3)		ROM_SI	ROM_SO	ROM_SCLK	STRAP0	STRAP1	STRAP2	STRAP3	STRAP4
N14P_GV2	Samsung 1GHZ	K4W2G1646E-BC1A	0x7	PU 4.99K	PU 4.99K	PU 45.3K	PD 4.99K	PD 15K	PD 4.99K	PU 45.3K
		128M x 16	PD 45.3K							
	Micron 1GMHz	MT41J128M16JT-093G:K	0x5							
		128M x 16	PD 30.1K							
	Hynix 1GMHz	H5TC2G63FFR-11C	0x4							
		128M x 16	PD 24.9K							
	Samsung 900MHz	K4W4G1646B-HC11	0x3							
		256M x 16	PD 20K							
Micron 900MHz	MT41K256M16HA-107G:E	0x1								
	256M x 16	PD 10K								

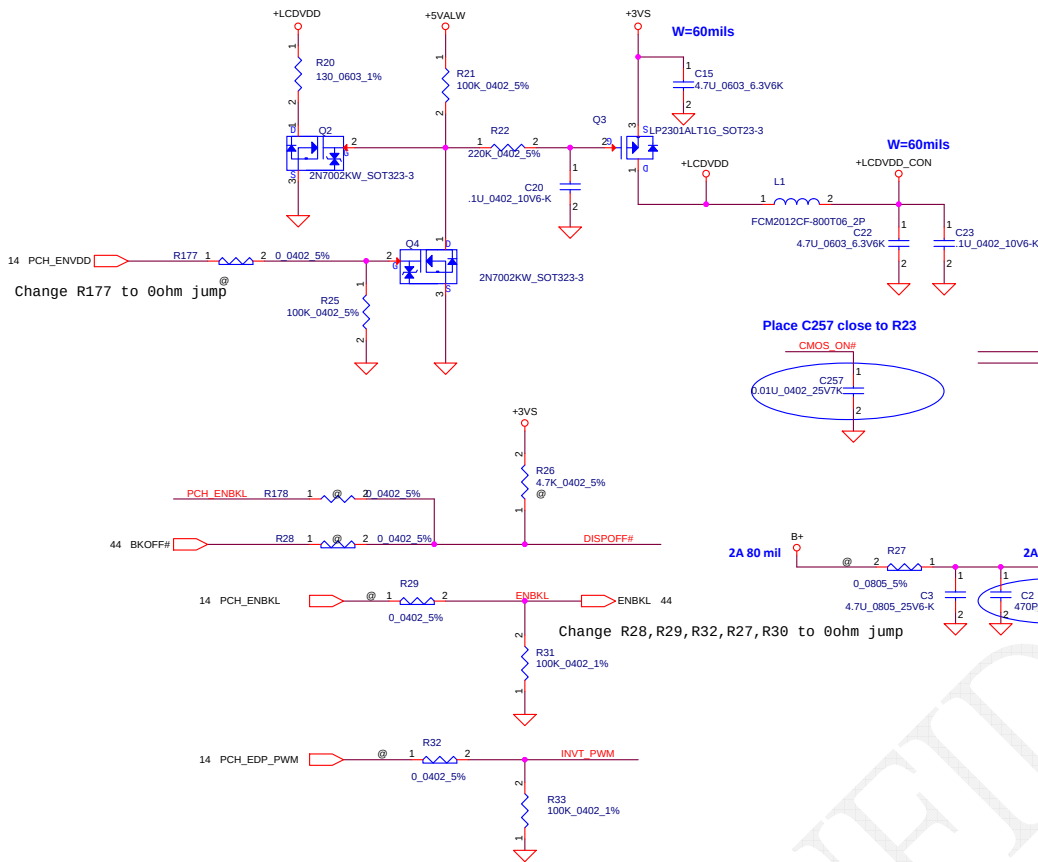
VRAM	X76	VRAM P/N
Samsung	X76409JVL01	SA00005SH10
	X76409JVL51 (1G 32Mx16)	
Micron	X76409JVL02	SA00005M100
	X76409JVL02 (2G 64Mx32)	
Hynix		

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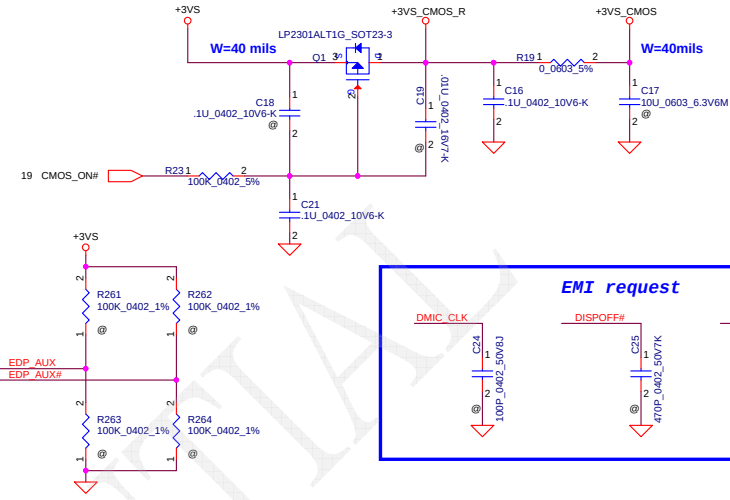
CONFIDENTIAL

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Issued Date	2012/12/14	Deciphered Date	2012/12/21	XDP_CONN			
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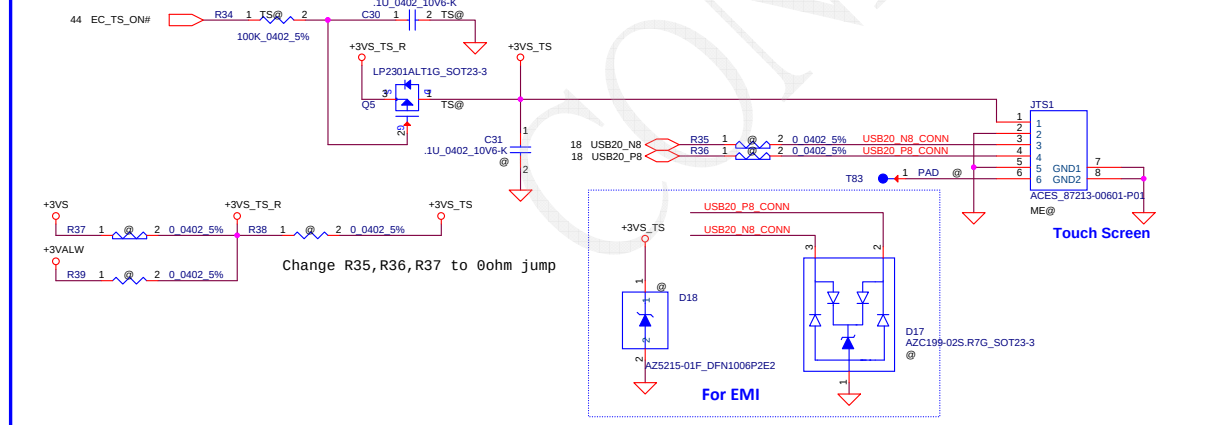
LCD POWER CIRCUIT

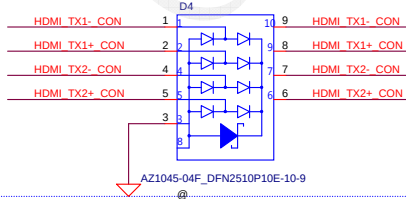
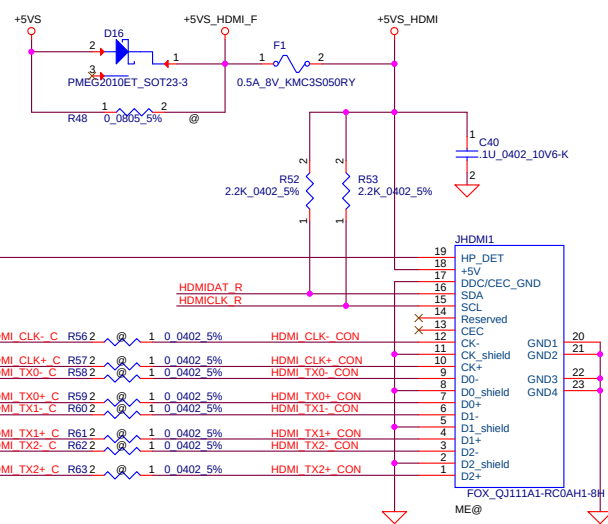
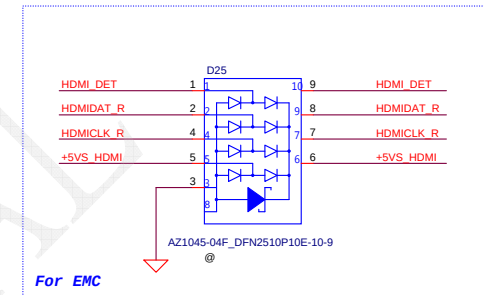


CMOS Camera

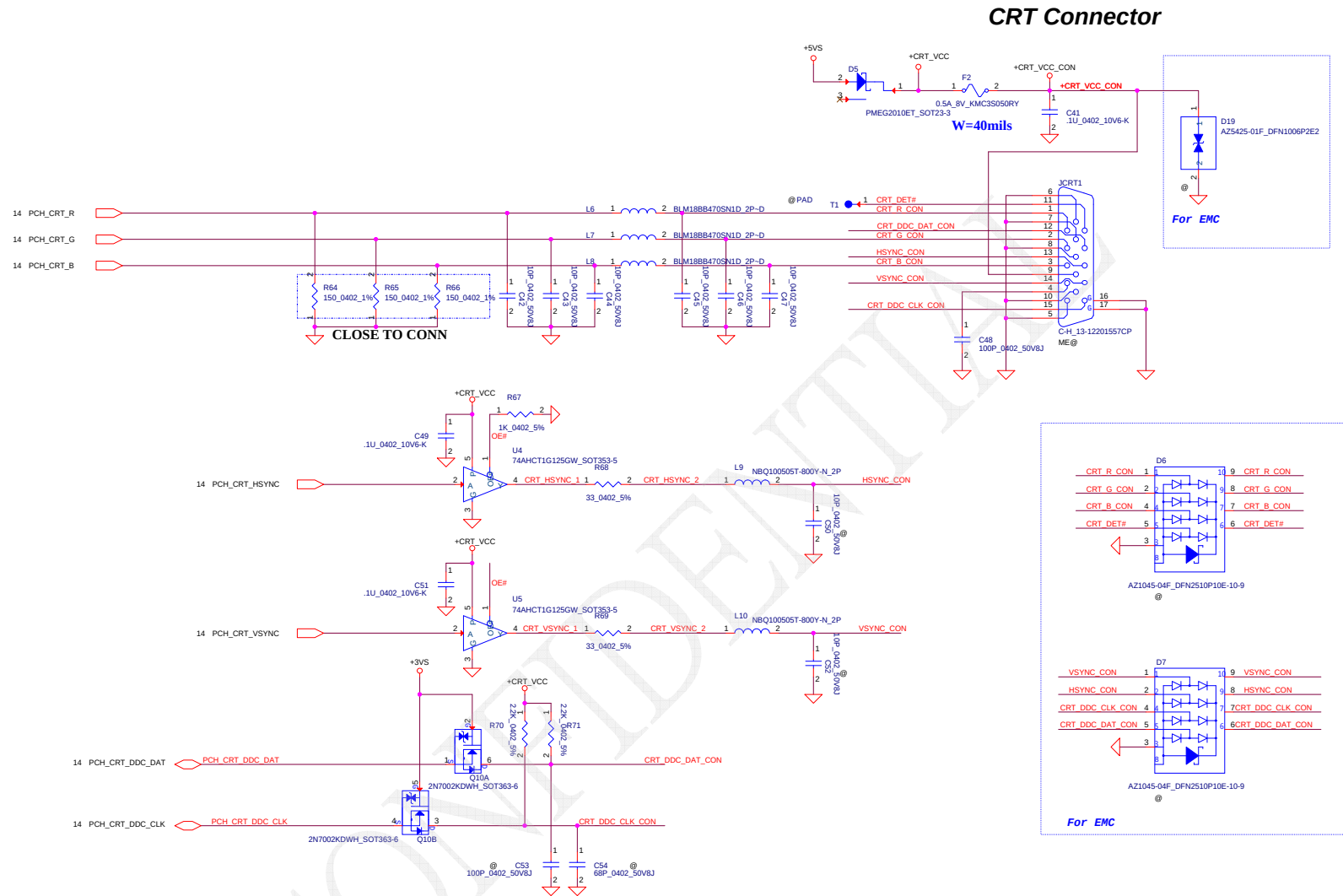


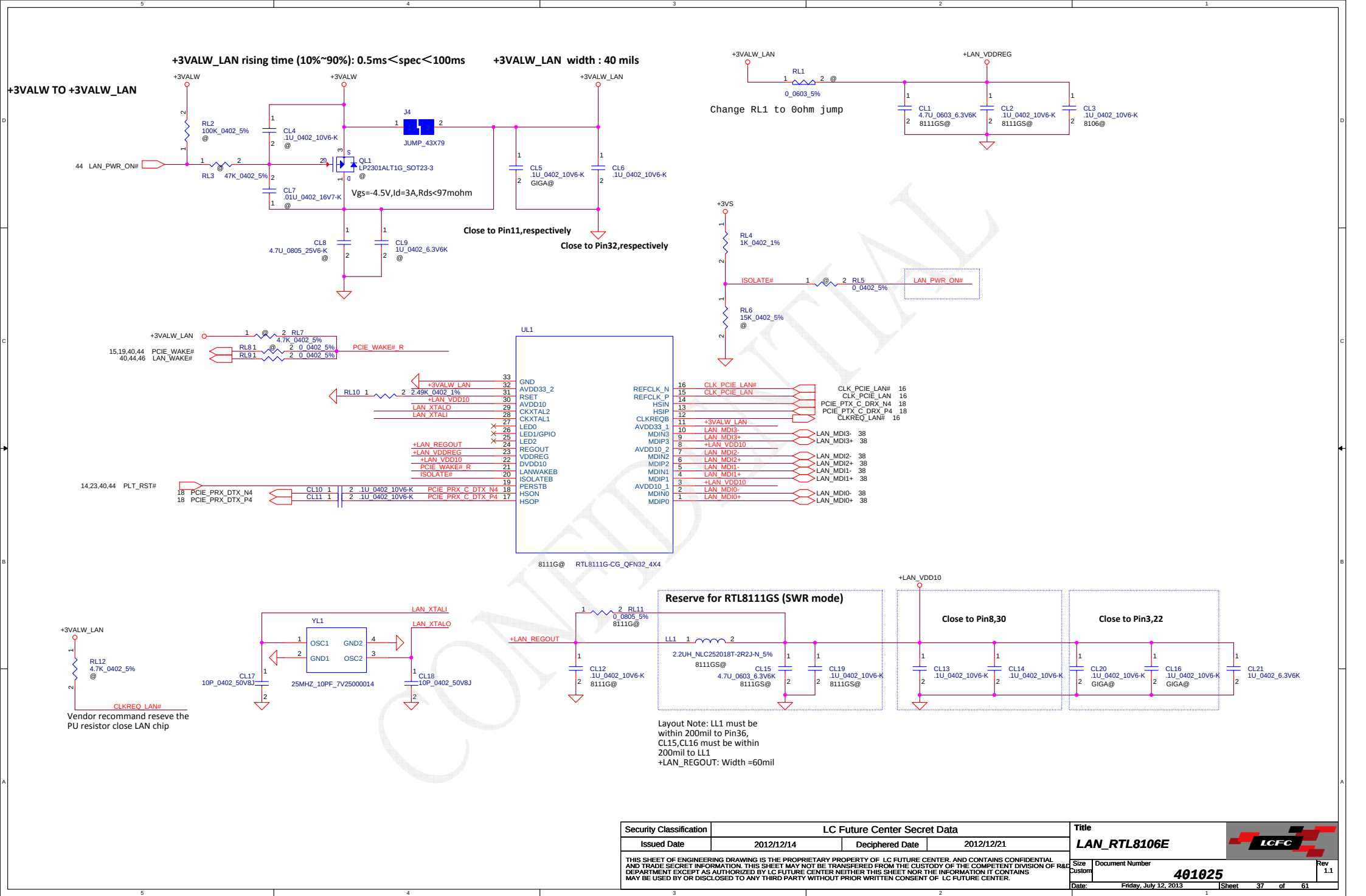
Touch Screen



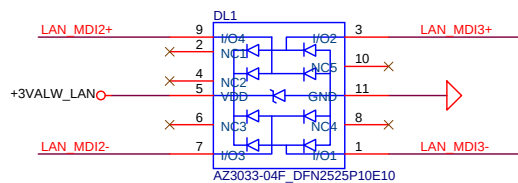


Title			
HDMI_CONN			
Size	Document Number	401025	
Custom			
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		Rev 1.1	

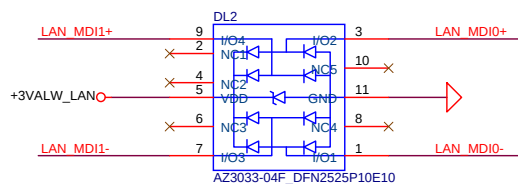




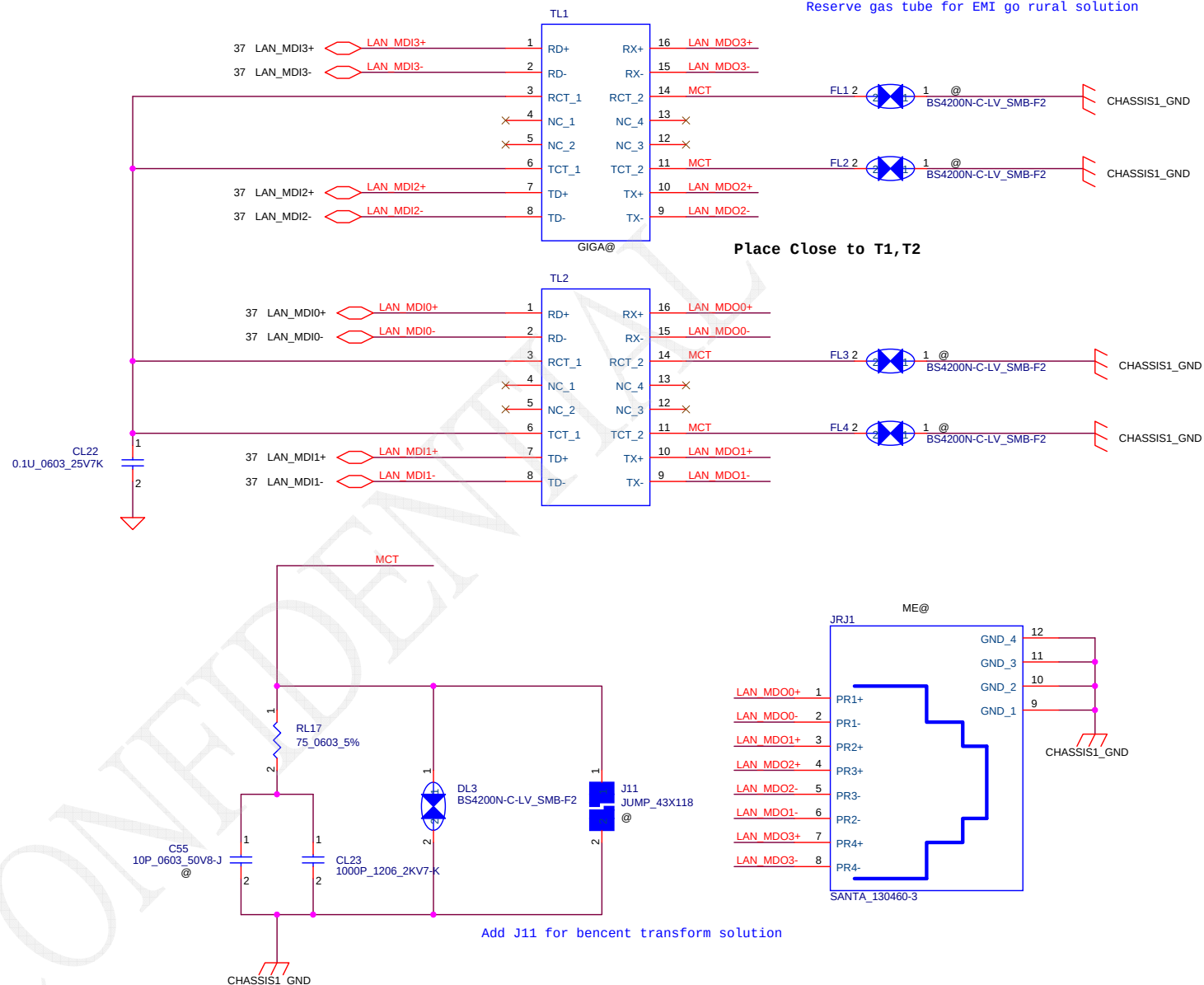
DL1/DL2
1'S PN:SC300003M00



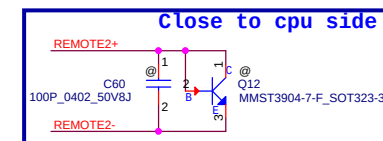
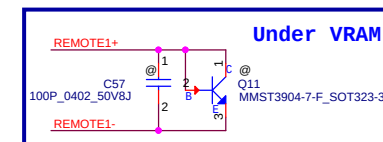
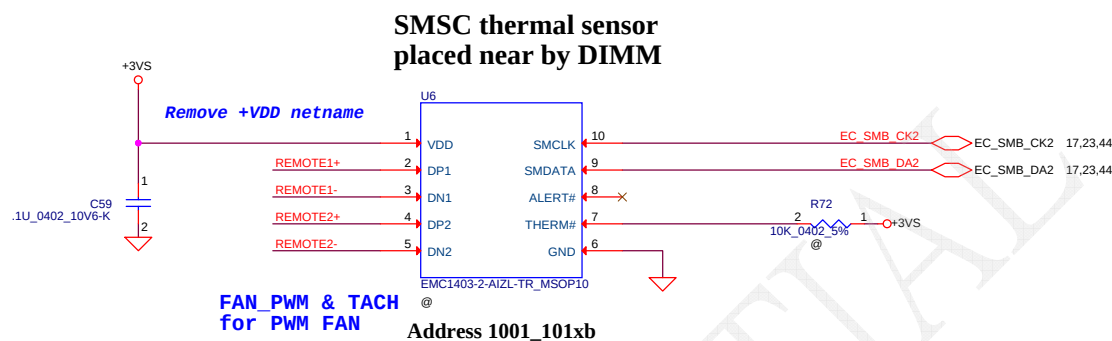
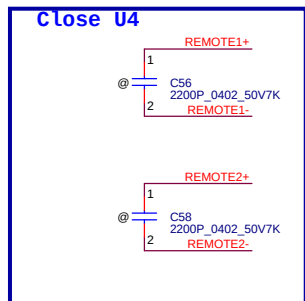
Place Close to TL1



Place Close to TL2

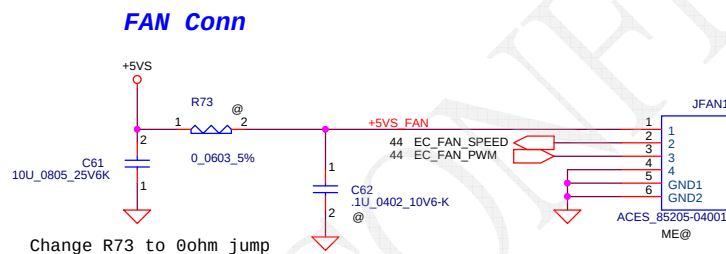


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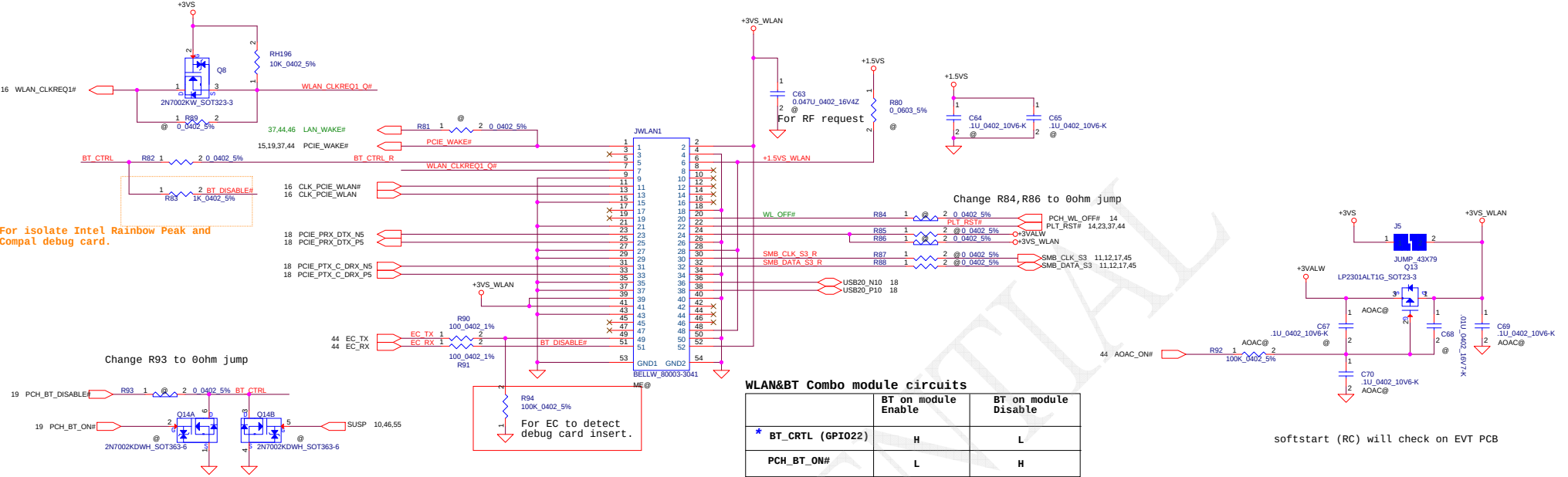
internal pull up 1.2K to 1.5V
R for initial thermal
shutdown temp

REMOTE2+/- :
Trace width/space:10/10 mil
Trace length:<8"

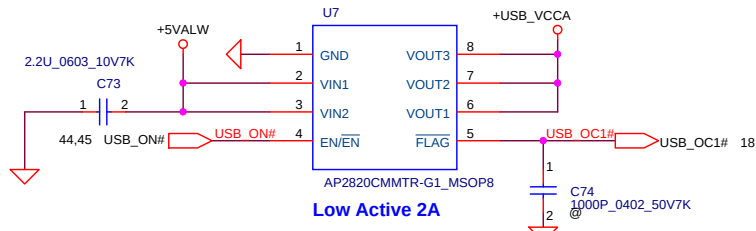


Security Classification		LC Future Center Secret Data		Title	
Issued Date	2012/12/14	Deciphered Date	2012/12/21	VGA Thermal sensor/FAN CONN	
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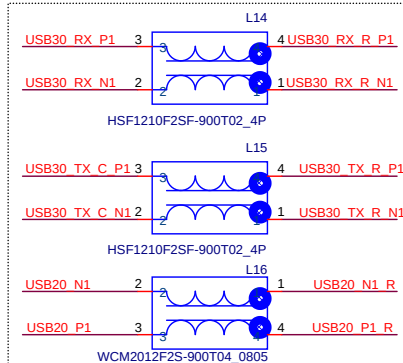
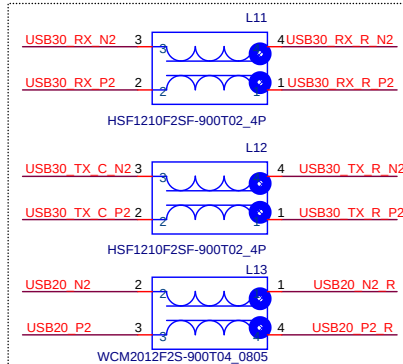
Mini-Express Card(WLAN/WiMAX)



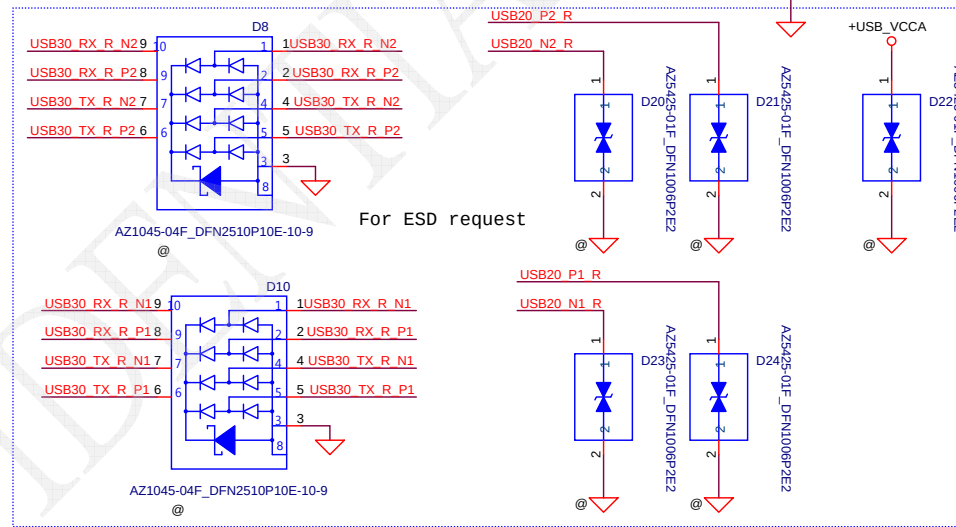
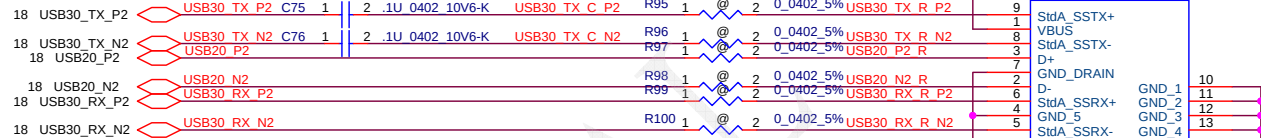
LEFT SIDE USB3.0 PORT X2



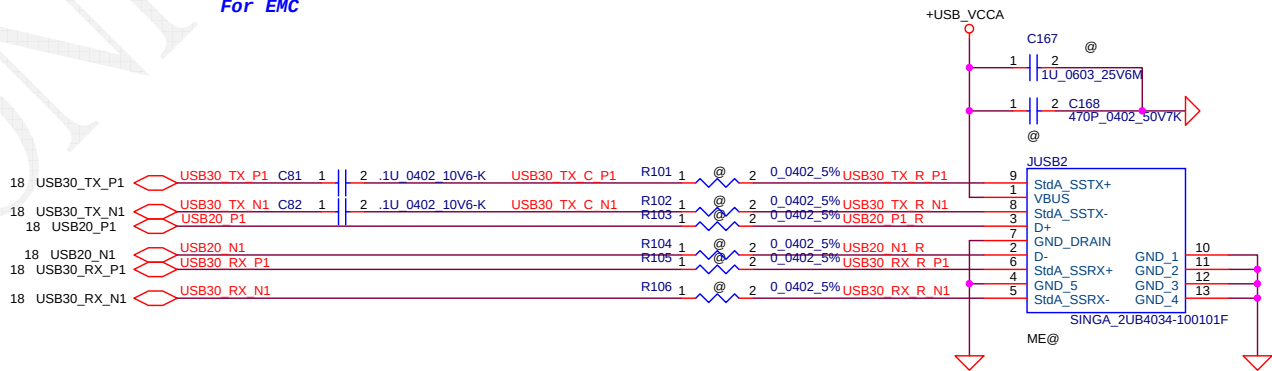
For EMI request
 USB2.0 choke --> SM070001S0J
 USB3.0 Choke --> SM070001S0J



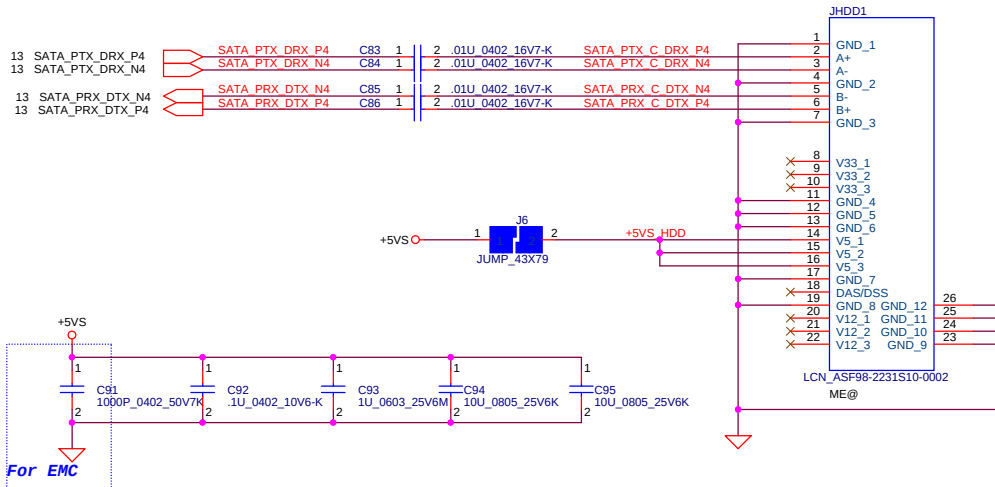
For EMC



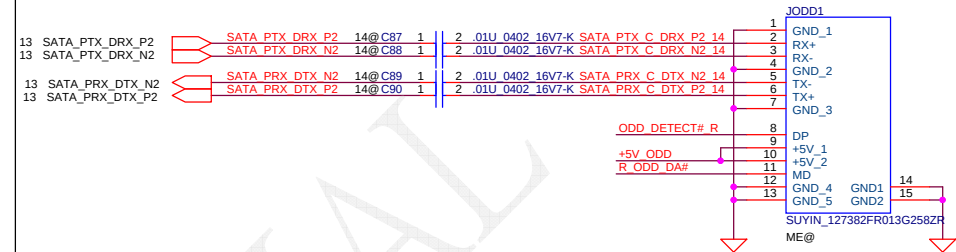
For EMC



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2012/12/14				2012/12/21				USB 3.0 PORT (LEFT)			
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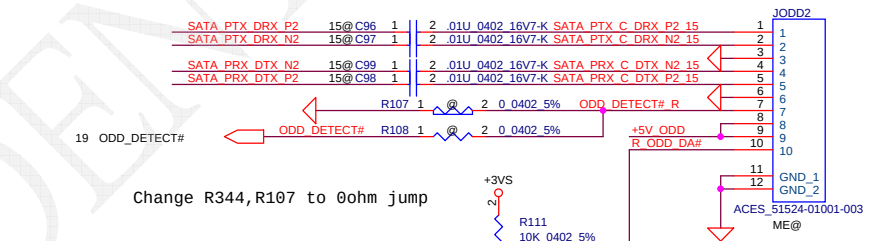
SATA HDD Conn.

**FOR 14"
SATA ODD Conn.**

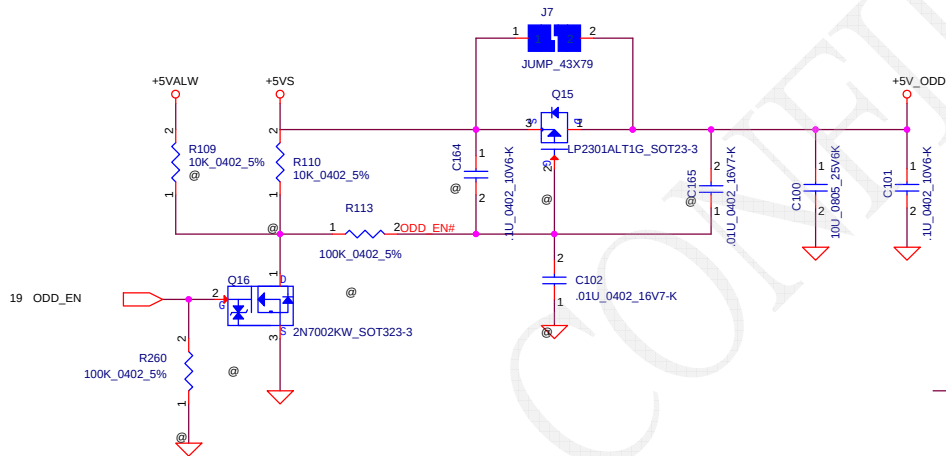
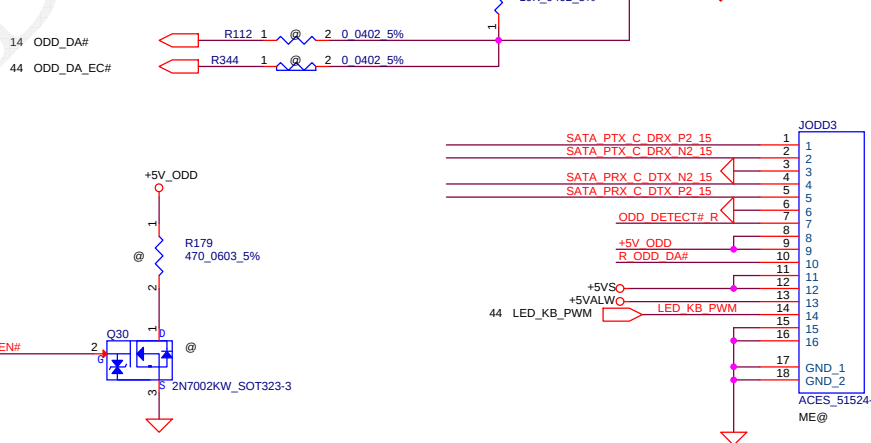


FOR 15"

Co-lay SATA ODD FFC Conn JODD2& JODD3.

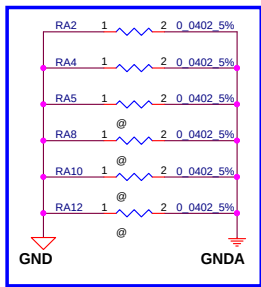


Change R344,R107 to 0ohm jump



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Title			
HDD/ODD CONN			
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The diagram shows two resistors, RA13 and RA14, connected in parallel between the +1.5VS supply and the AVDD2 pin. Both resistors are 0.0402 Ohms with a 5% tolerance.

EC BEEP

44 BEEP#

CA14

1 2

.1U_0402_10V6-K

PC_BEEP1

PCH BEEP

13 PCH_SPKR

CA17

1 2

.1U_0402_10V6-K

1 RA22 2

33_0402_5%

RA23 1K_0402_5%

PC_BEEP

RA29 1 2 0 0402_5%

HDA_RST_AUDIO#

DA1

1 2

RB751V-40_SOD323-2

PDB

RA24 is but code

_MUTE#

EC_MUTE#

DA2

1 2

RB751V-40_SOD323-2

AND GATE

RA30 1 2 0 0402_5%

RA40 100K_0402_5%

When SPKR_MUTE# is high, trun on AMP.
When SPKR_MUTE# is low, trun off AMP.

+MIC2_VREF0_R

RA35 2.2K_0402_5%

RA37 1 2

1K_0402_0.5%

RA38 22K_0402_5%

RA39 22K_0402_5%

COMBO-GPI

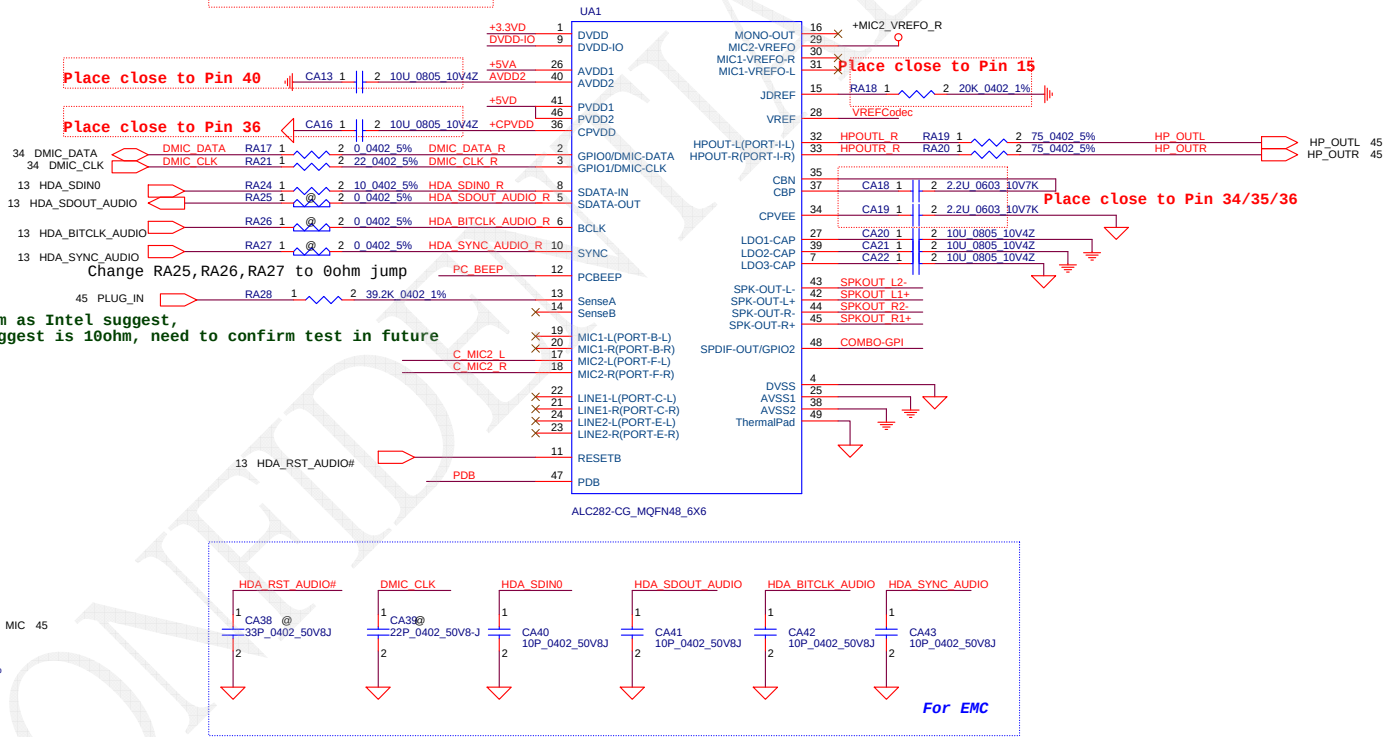
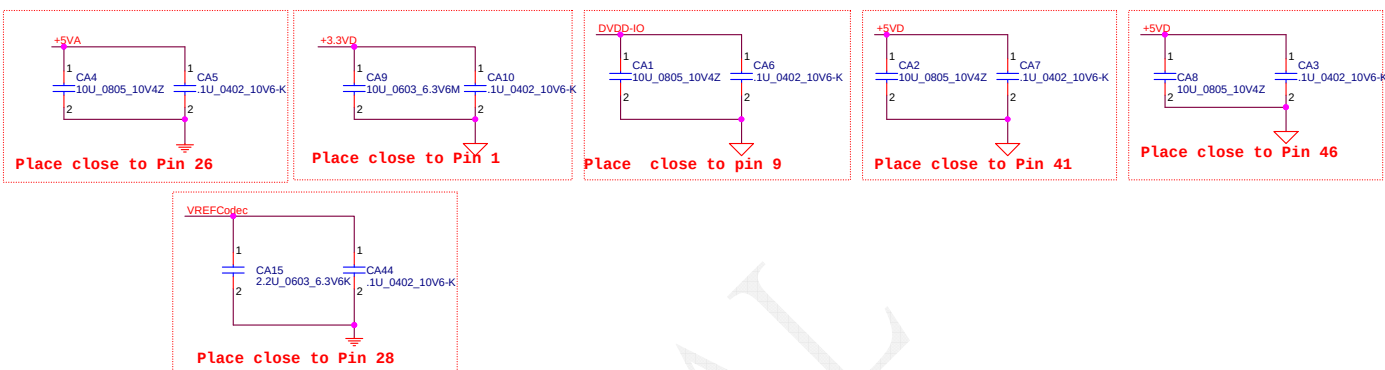
CA27 1 2 2.2U 0603 6.3V6K

C2 L

CA28 1 2 2.2U 0603 6.3V6K

C2 R

CA34 10U_0603_10V6K



SPKOUT L1+ RA31 1 2 0 0603 5%
SPKOUT L2- RA32 1 2 0 0603 5%
SPKOUT R1+ RA33 1 2 0 0603 5%
SPKOUT R2- RA34 1 2 0 0603 5%

to 0ohm jump

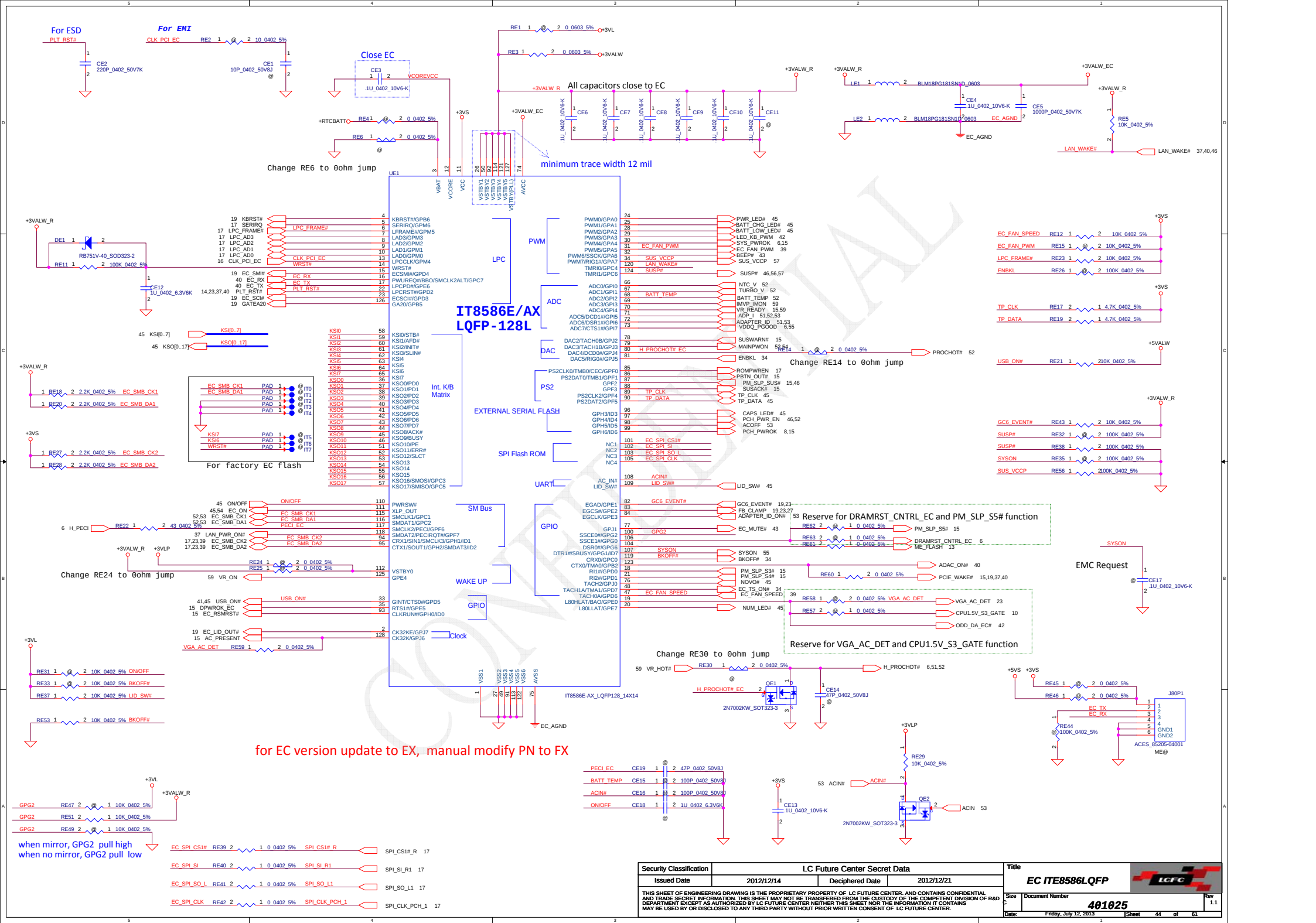
For EMC

Ca23 1 2 1000p 0402 50V7K
Ca24 1 2 1000p 0402 50V7K
Ca25 1 2 1000p 0402 50V7K
Ca26 1 2 1000p 0402 50V7K
Ca26 1 2 1000p 0402 50V7K

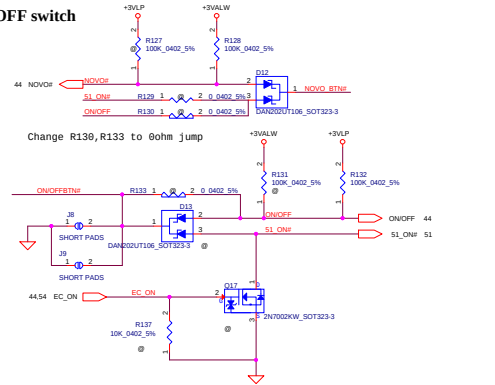
SPK L1 1
SPK L2 2
SPK R1 3
SPK R2 4

ACES_88231-04001
ME@

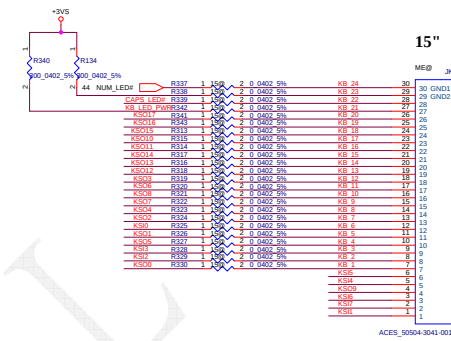
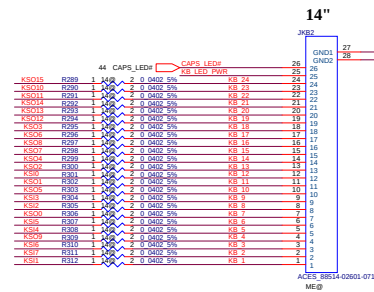
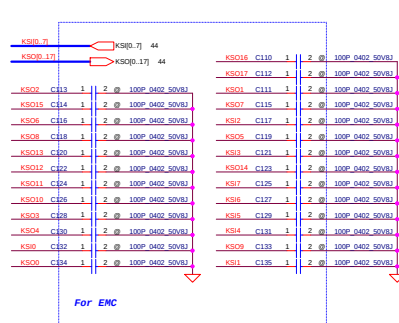
Title Audio Codec_ALC282			
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ON/OFF switch

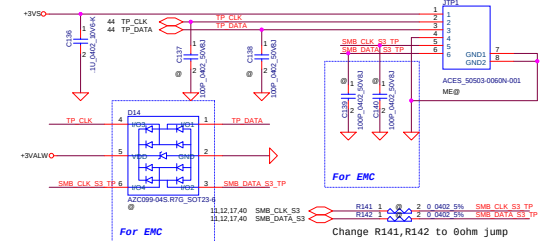


K/B Connector

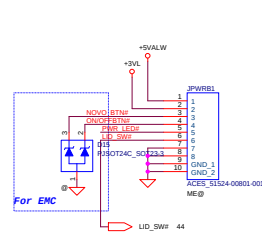


KB_1	K511_14	K506_15
KB_2	K517_14	K512_15
KB_3	K516_14	K513_15
KB_4	K509_14	K505_15
KB_5	K514_14	K501_15
KB_6	K515_14	K510_15
KB_7	K508_14	K502_15
KB_8	K512_14	K504_15
KB_9	K513_14	K507_15
KB_10	K505_14	K508_15
KB_11	K501_14	K506_15
KB_12	K502_14	K503_15
KB_13	K507_14	K501_15
KB_14	K504_14	K501_15
KB_15	K507_14	K501_15
KB_16	K508_14	K501_14
KB_17	K506_14	K501_15
KB_18	K503_14	K501_15
KB_19	K501_14	K501_15
KB_20	K501_14	K501_15
KB_21	K501_14	KB_LED_PWR
KB_22	K501_14	CAPS_LED_15
KB_23	K501_14	VDD_15
KB_24	K501_14	NUM_LEDW_15

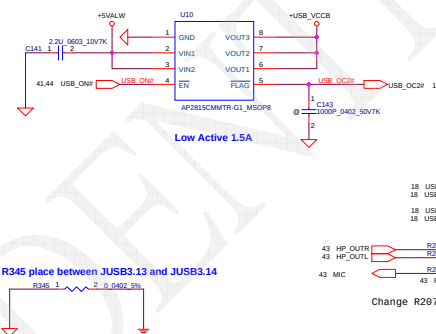
TP/B Connector



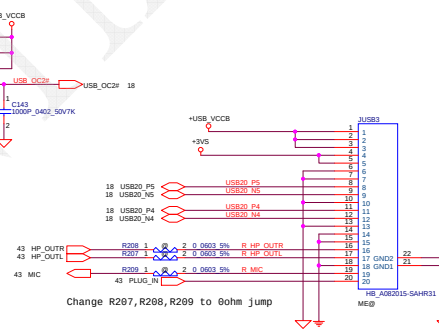
PWR/B Connector



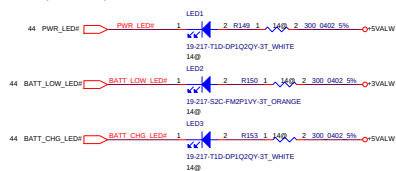
Right Side USB2.0 Port X 1 (USB/B)



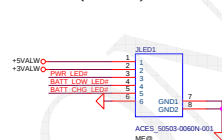
USB I/O Connector



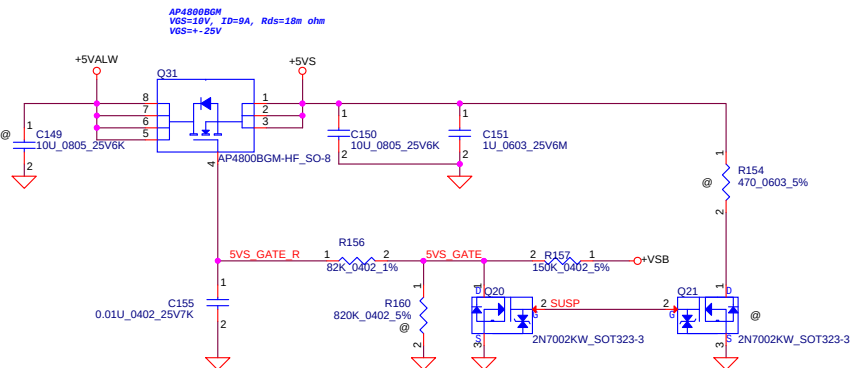
LED (For 14")



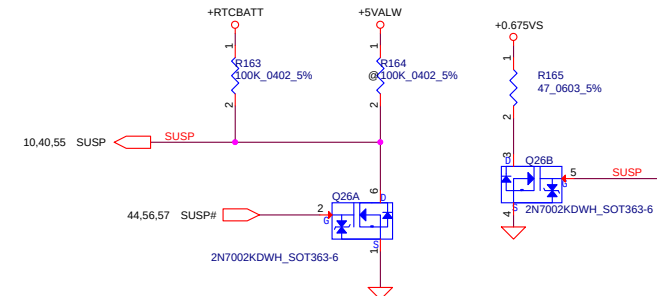
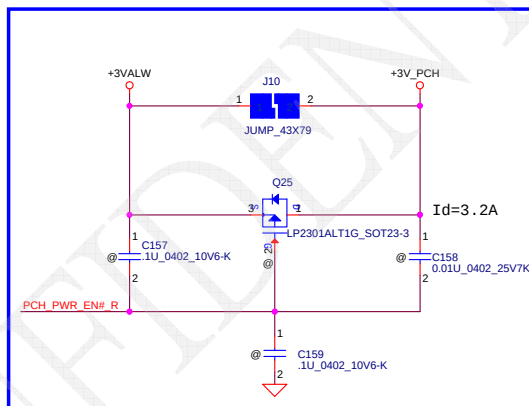
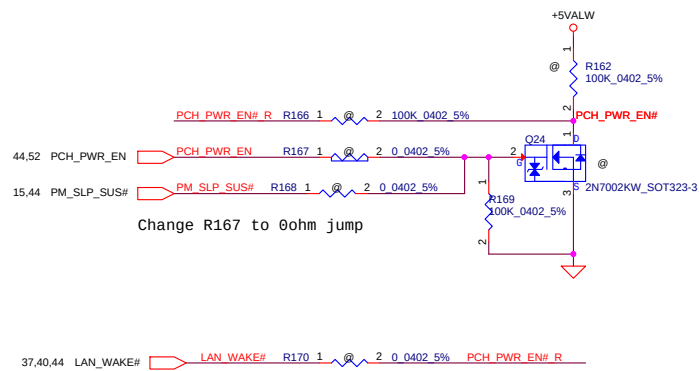
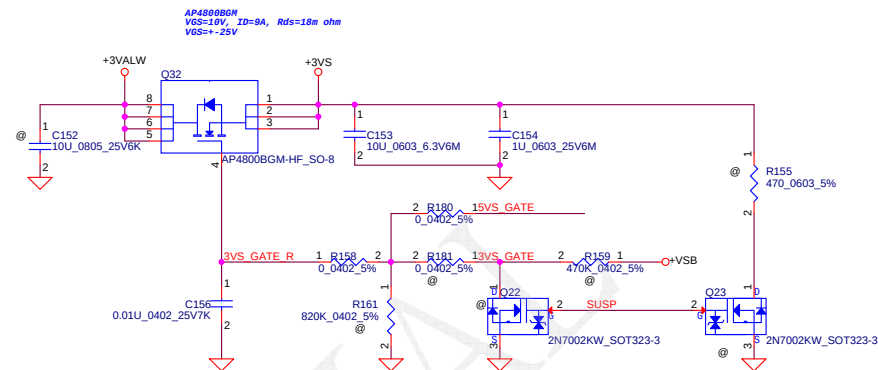
LED (For 15")



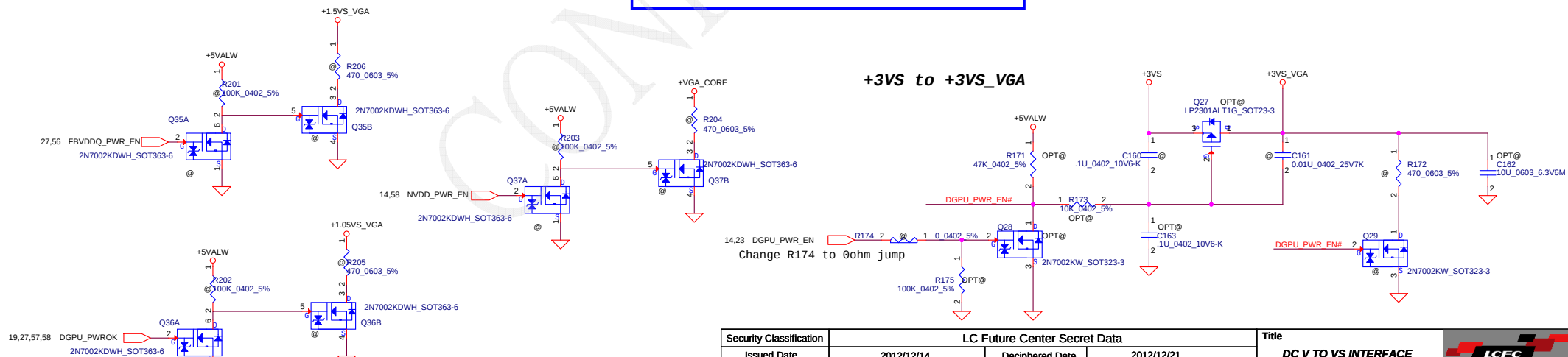
+5VALW to +5VS



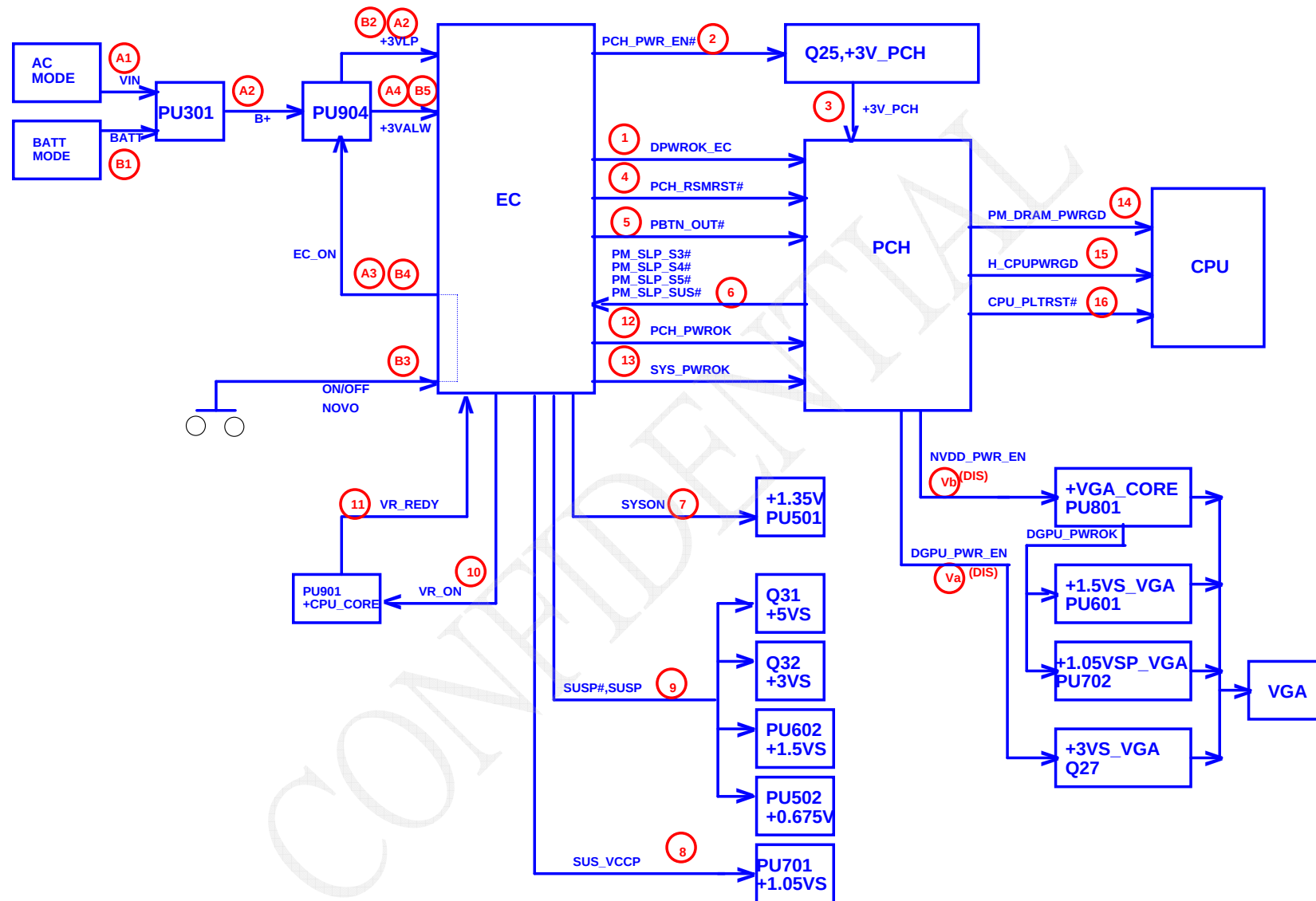
+3VALW to +3VS

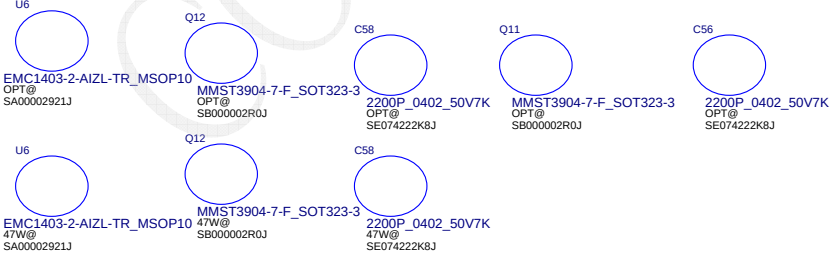
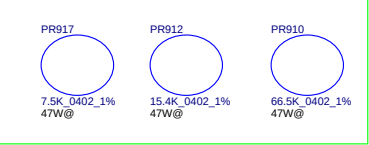
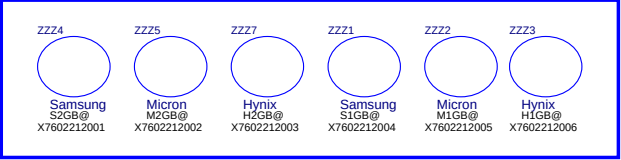
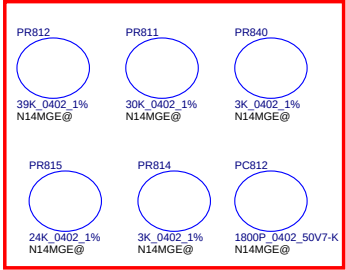
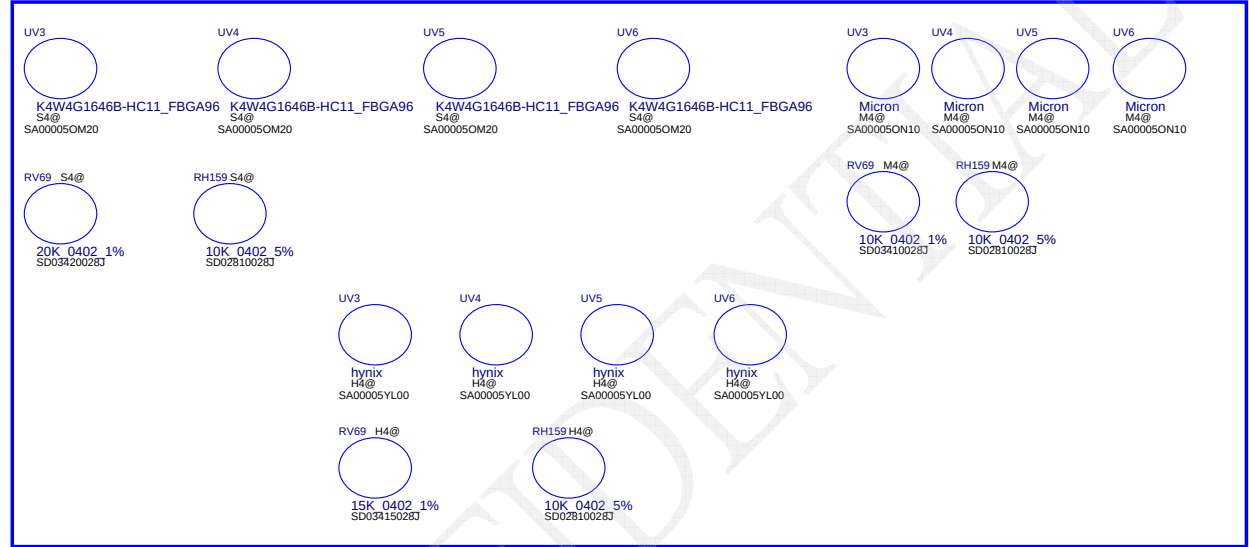
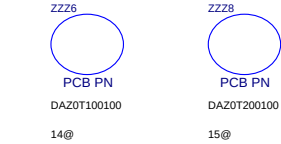
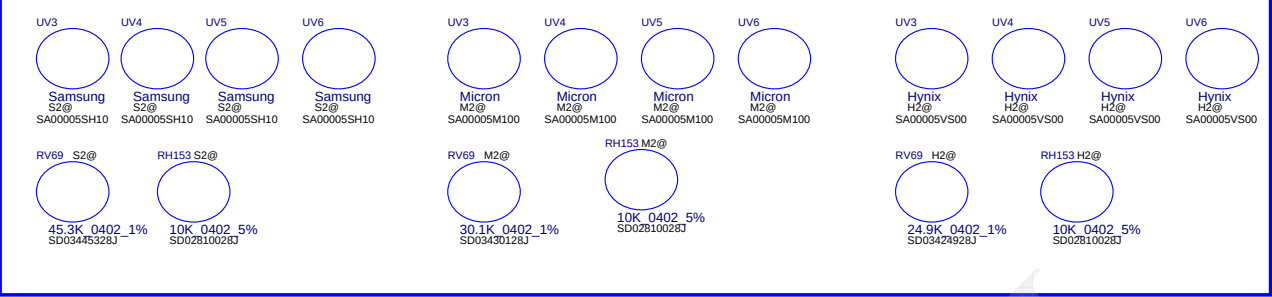
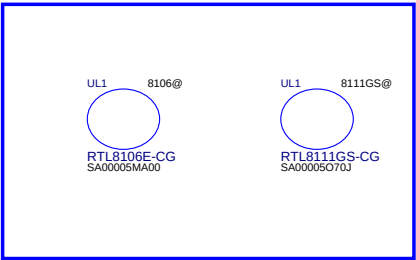


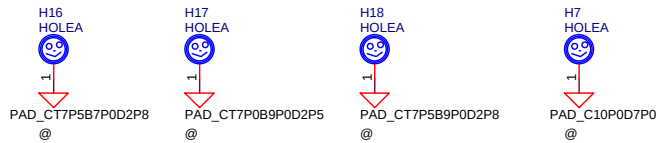
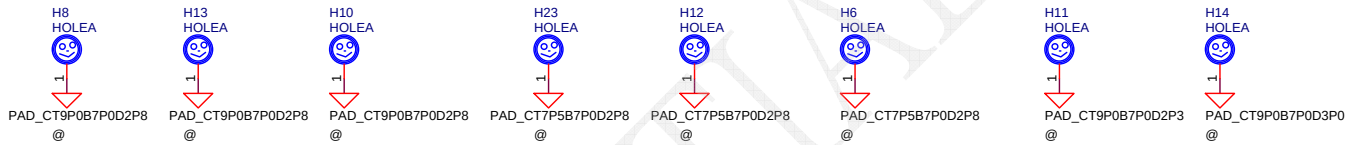
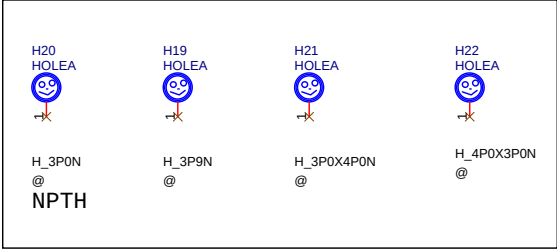
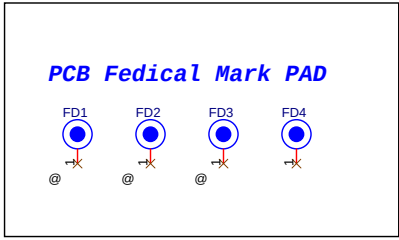
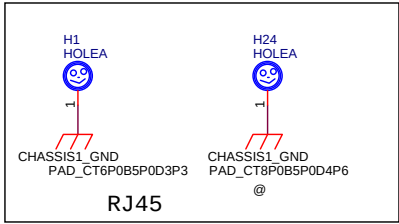
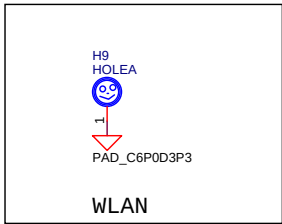
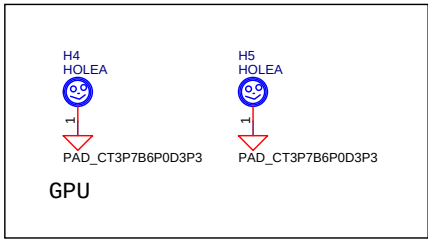
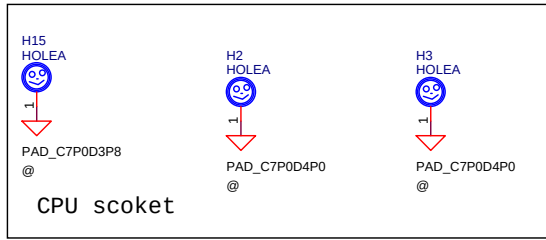
+3VS to +3VS_VGA



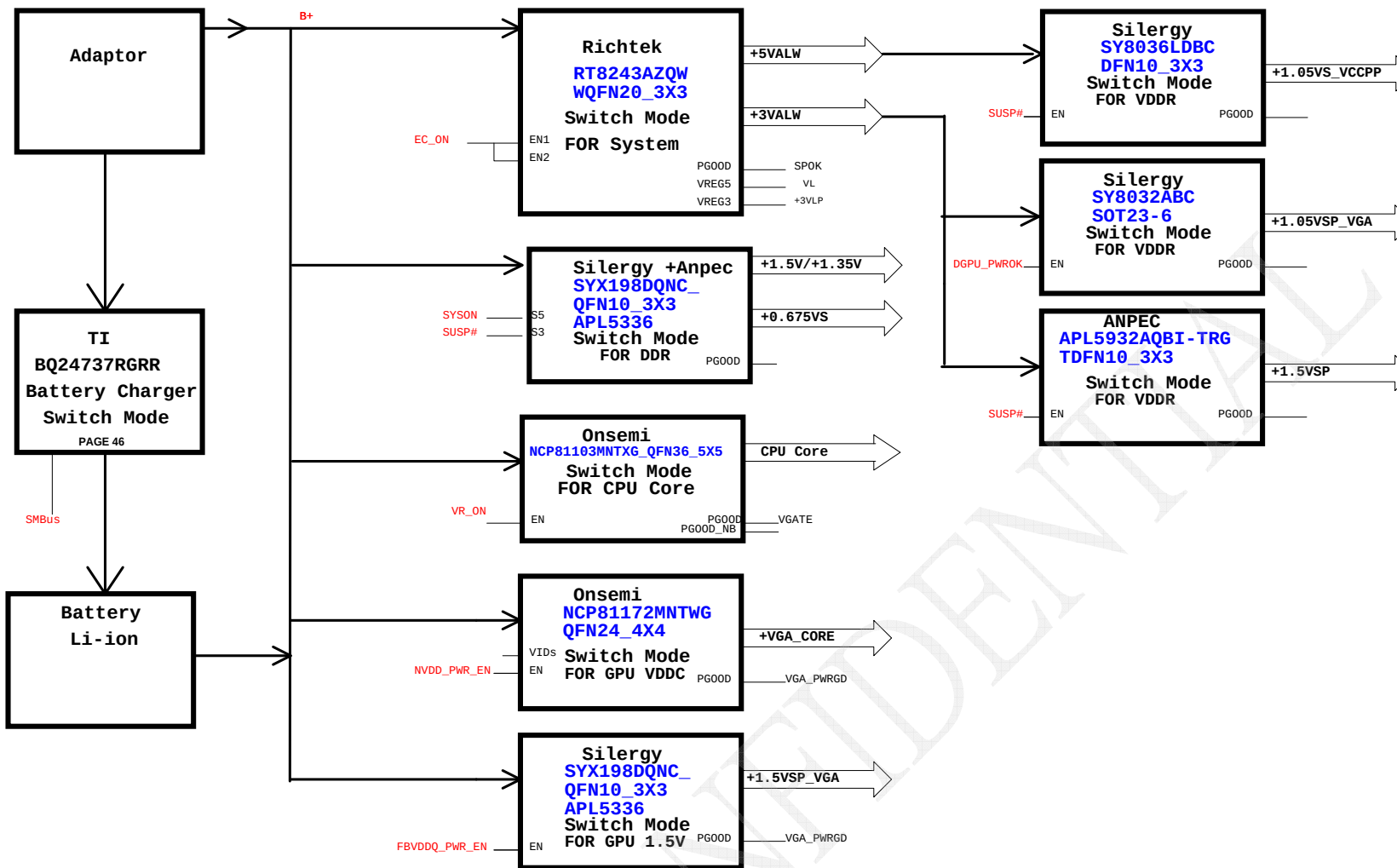
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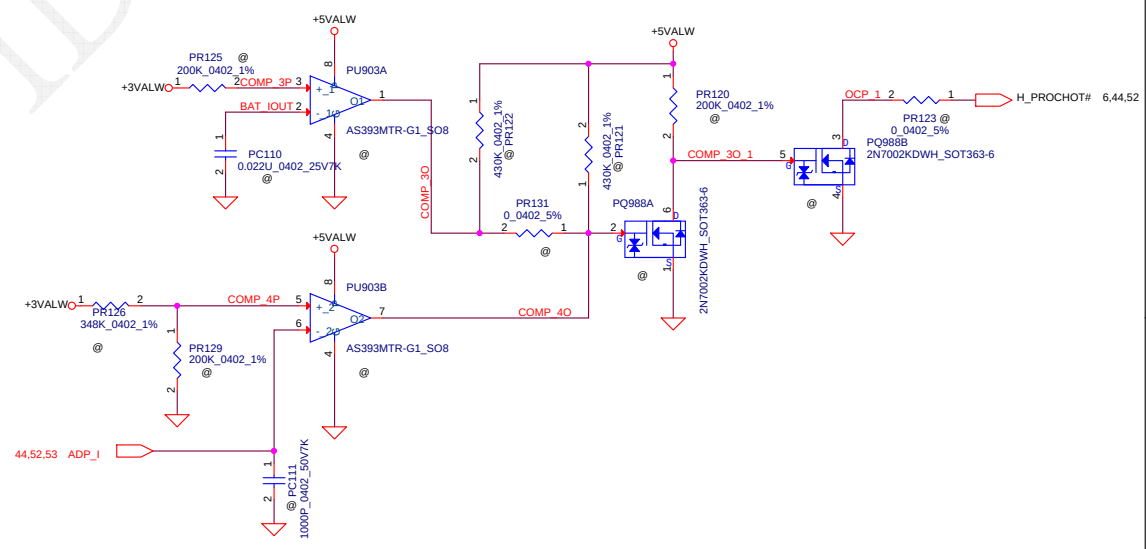
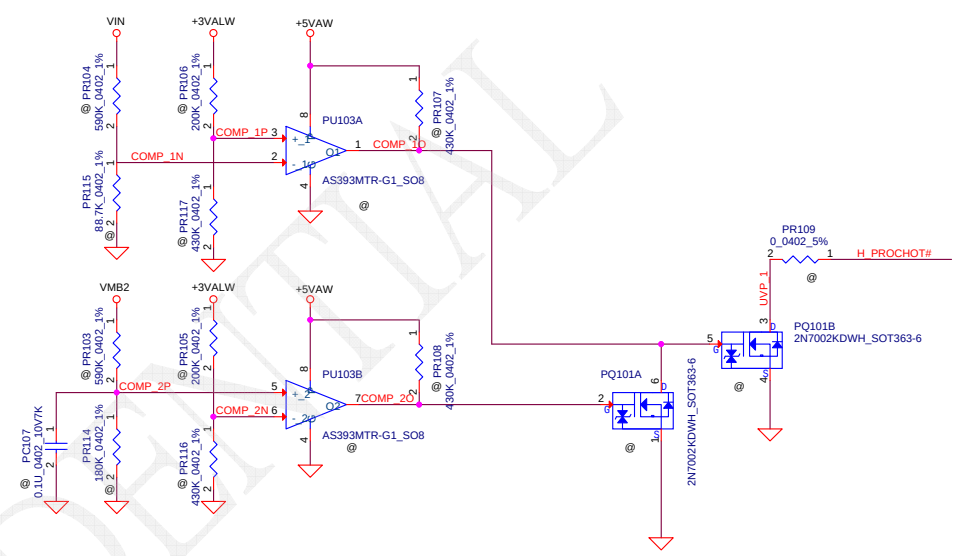
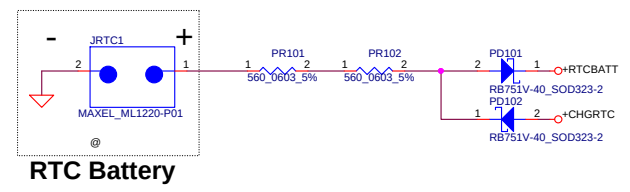




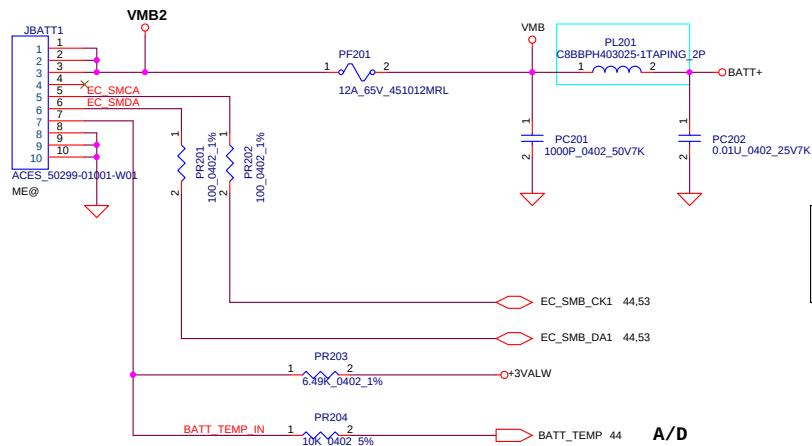


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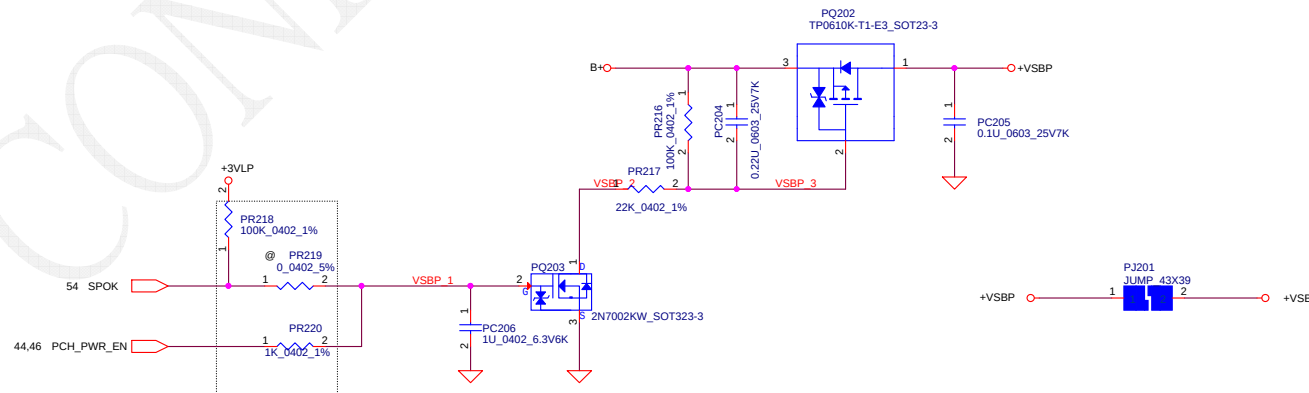
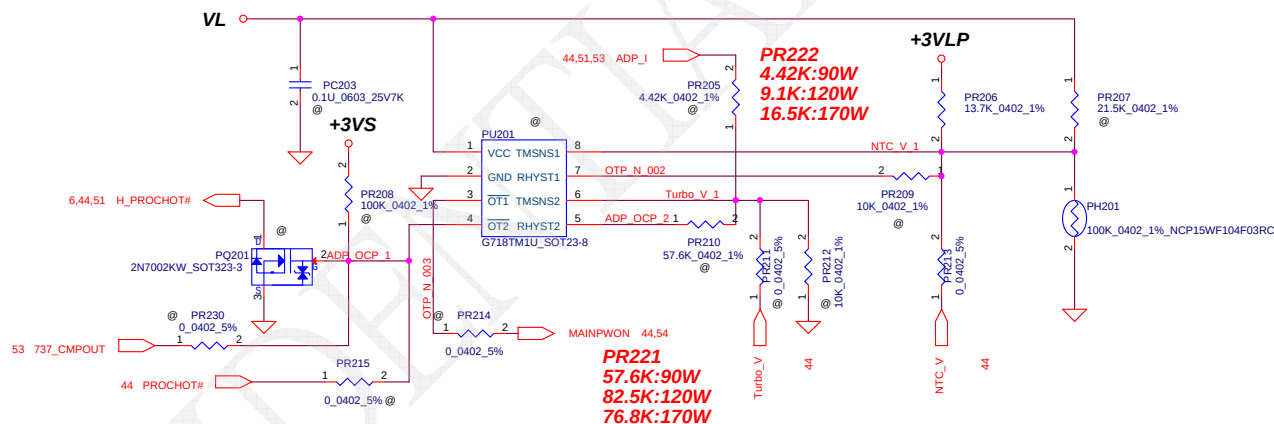


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Issued Date	2012/12/28	Deciphered Date	2012/12/28	DCIN / RTC charger			
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PH1 under CPU bottom side :
CPU thermal protection at 92+3 degree C
Recovery at 56 +3 degree C

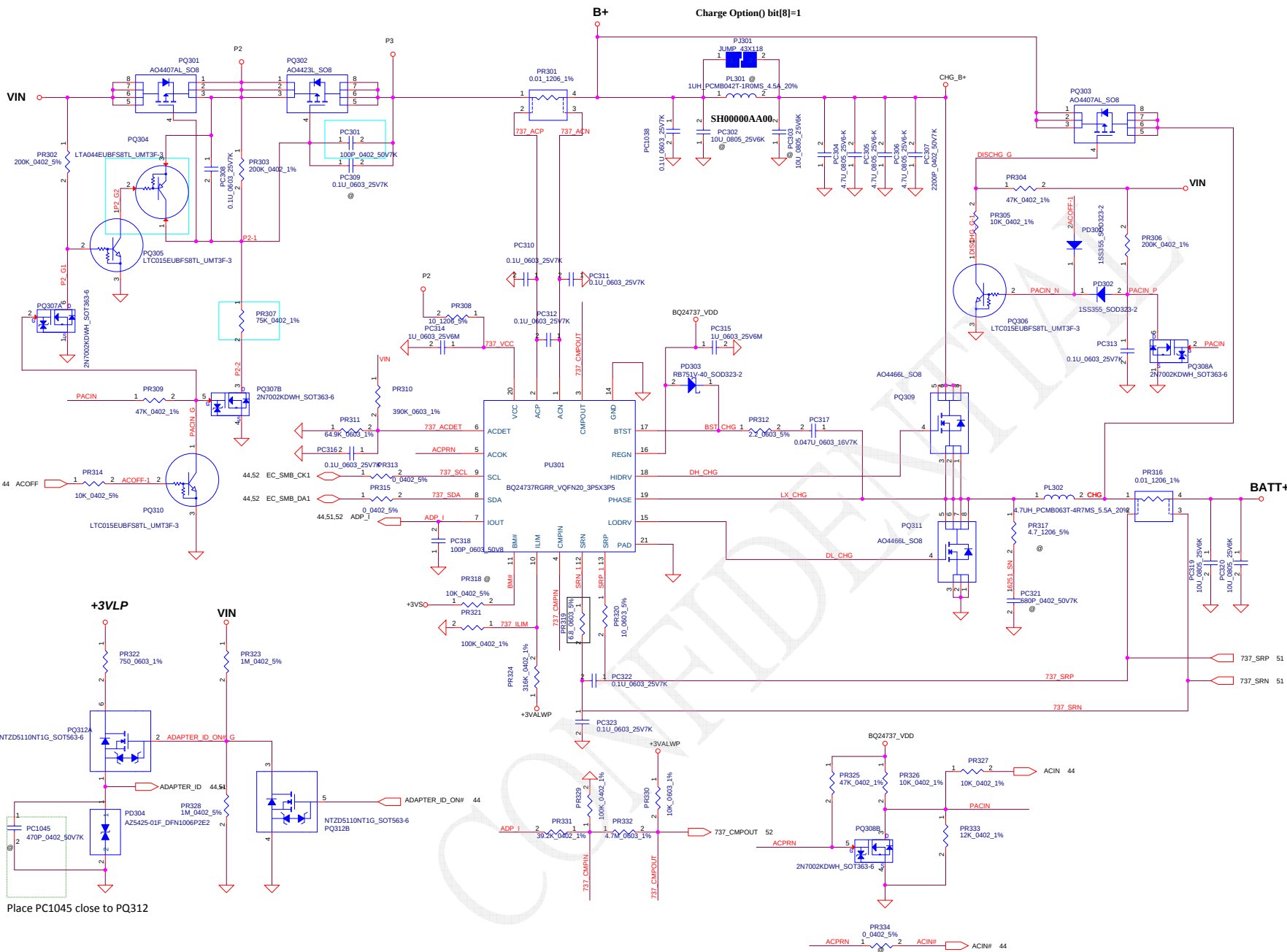
For KB930 --> Keep PU1 circuit
(Vth = 0.825V)
For KB9012 (Red square) --> Remove PU201 circuit, but keep PR206
PH201, PR205, PR211, PQ201, PR208, PR212



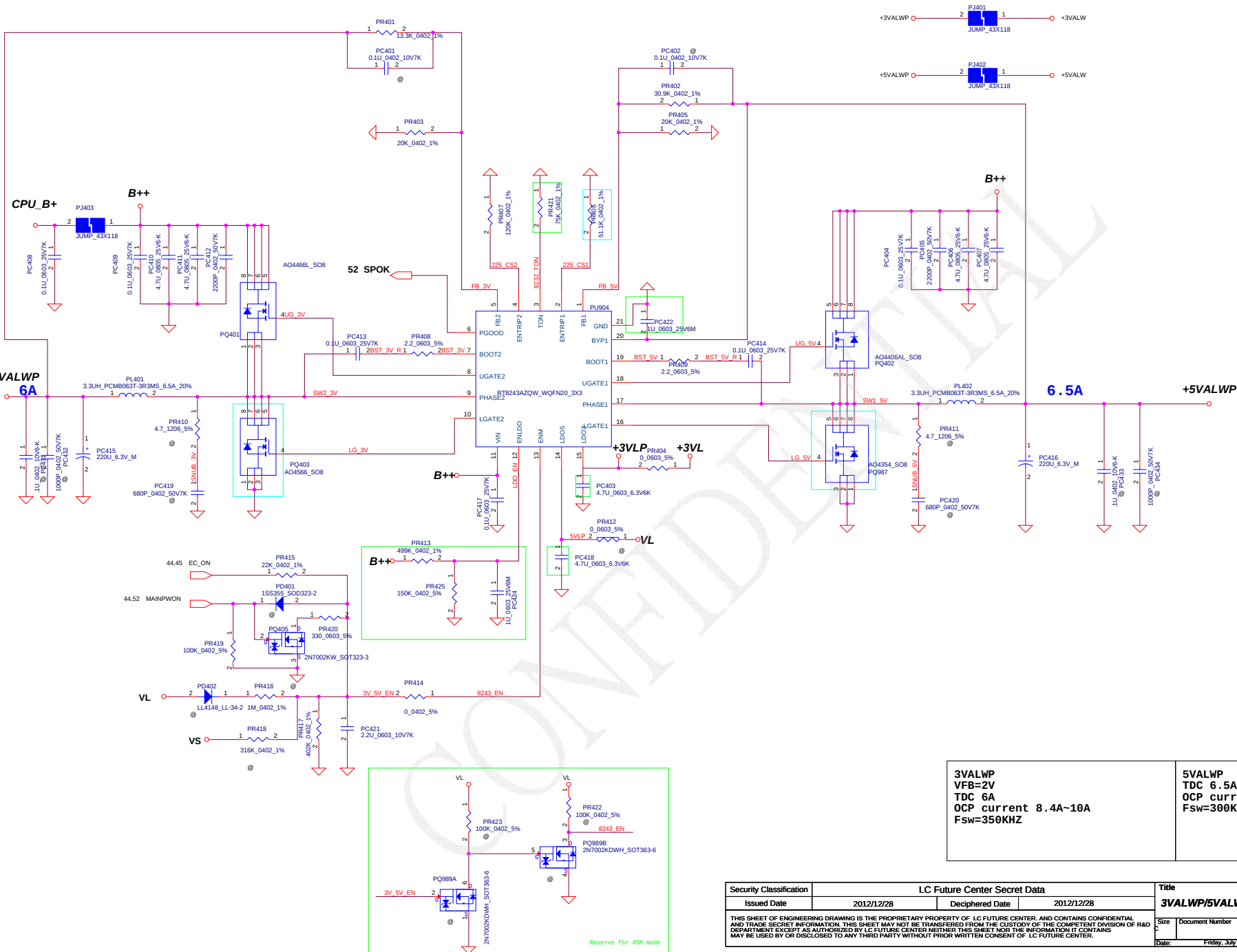
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Issued Date	2012/12/28	Deciphered Date
		2012/12/28

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Place PC1045 close to PQ312



3VALWP VFB=2V TDC 6A OCP current 8.4A-10A Fsw=350KHZ	5VALWP TDC 6.5A OCP current 10.3A-12A Fsw=300KHZ
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Security Classification	LC Future Center Secret Data		Title	3VALWP/5VALWP	
Issued Date	2012/12/28	Deciphered Date	2012/12/28	Size	Document Number
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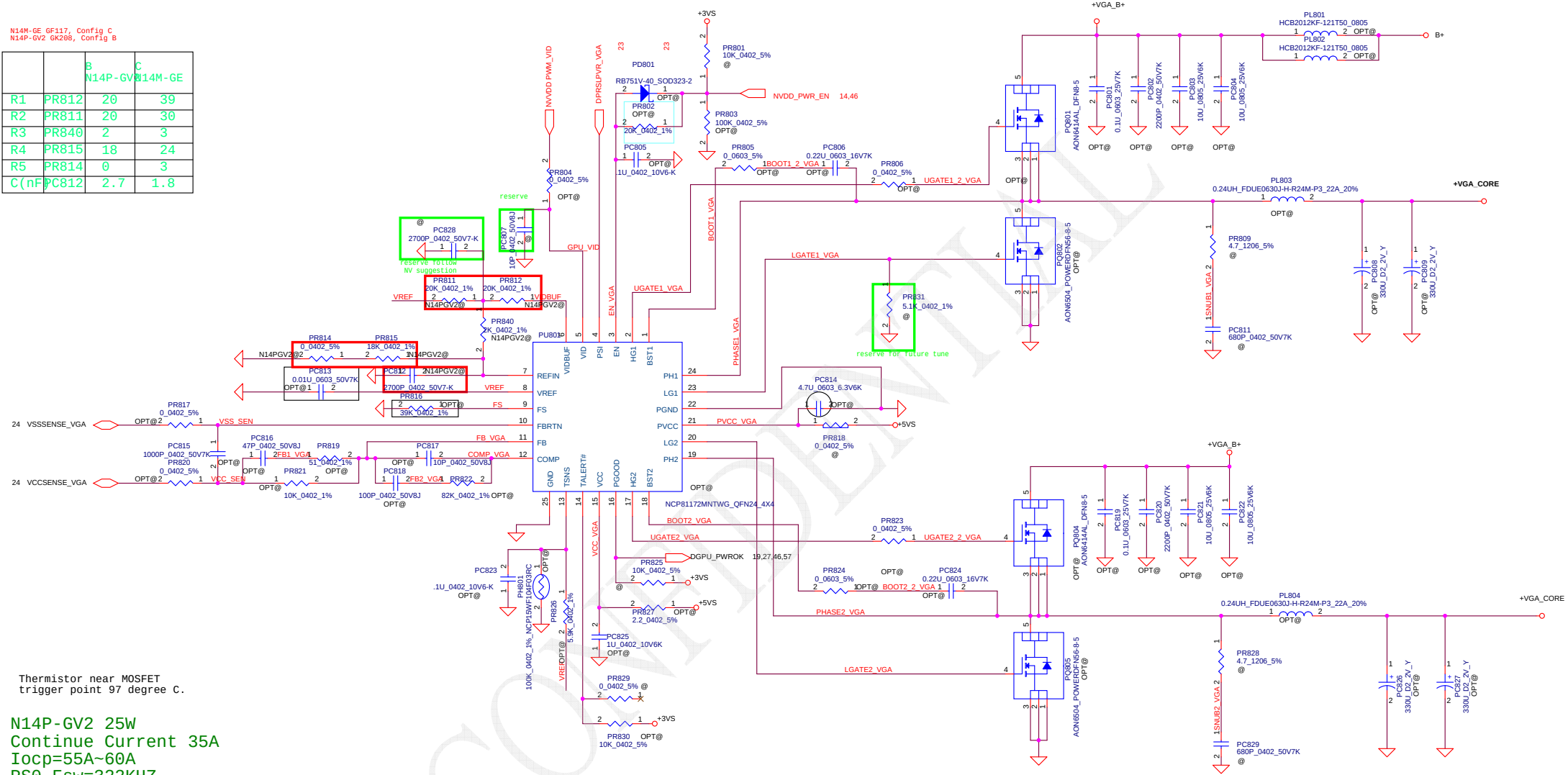


401025

Rev 1.1

N14M-GE 0F117, Config C
N14P-GV2 GK208, Config B

	B	C
	N14P-GV2	N14M-GE
R1	PR812	20
R2	PR811	20
R3	PR840	2
R4	PR815	18
R5	PR814	0
C(nF)	PC812	2.7
		1.8

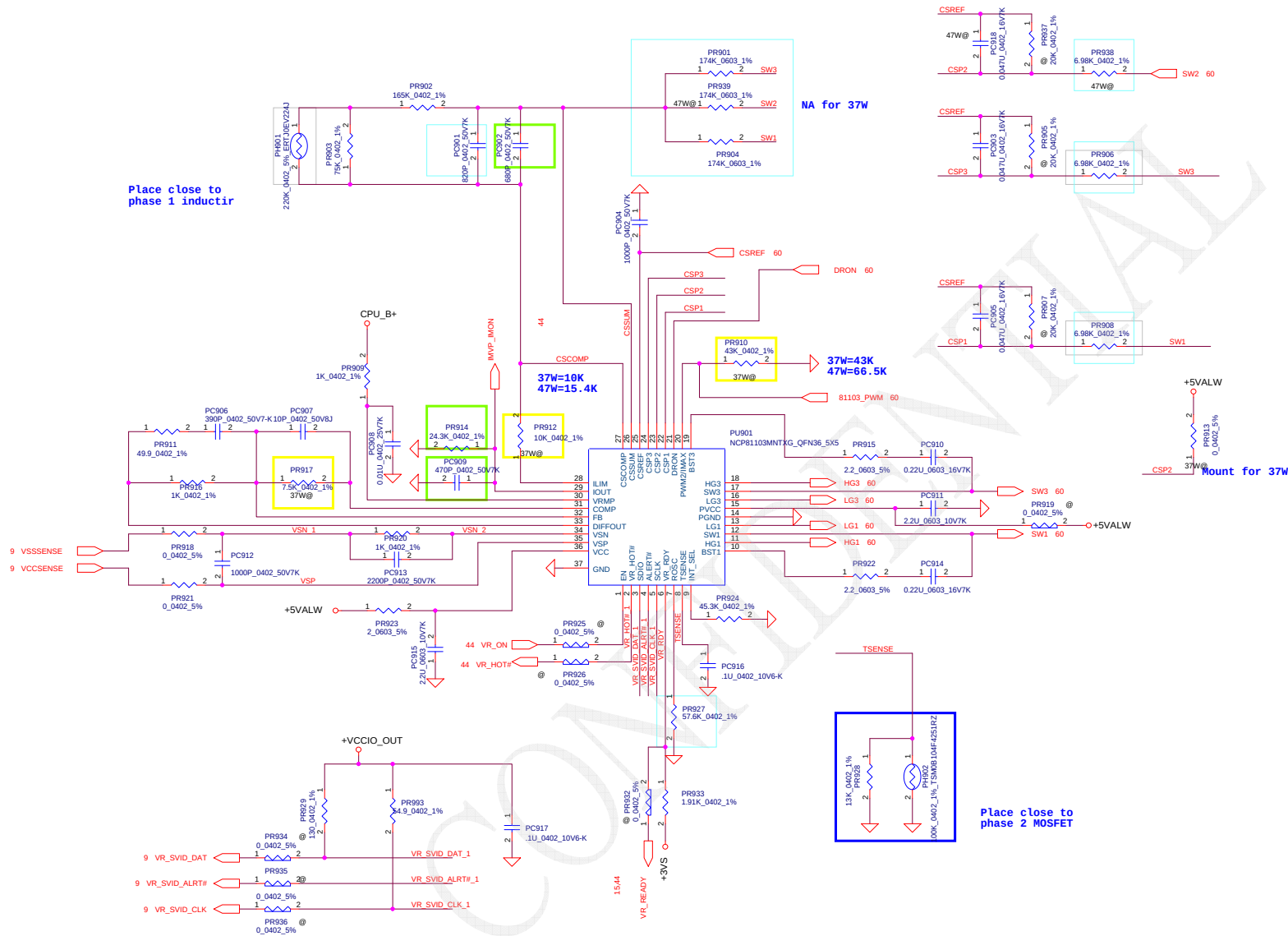


Thermistor near MOSFET
trigger point 97 degree C.

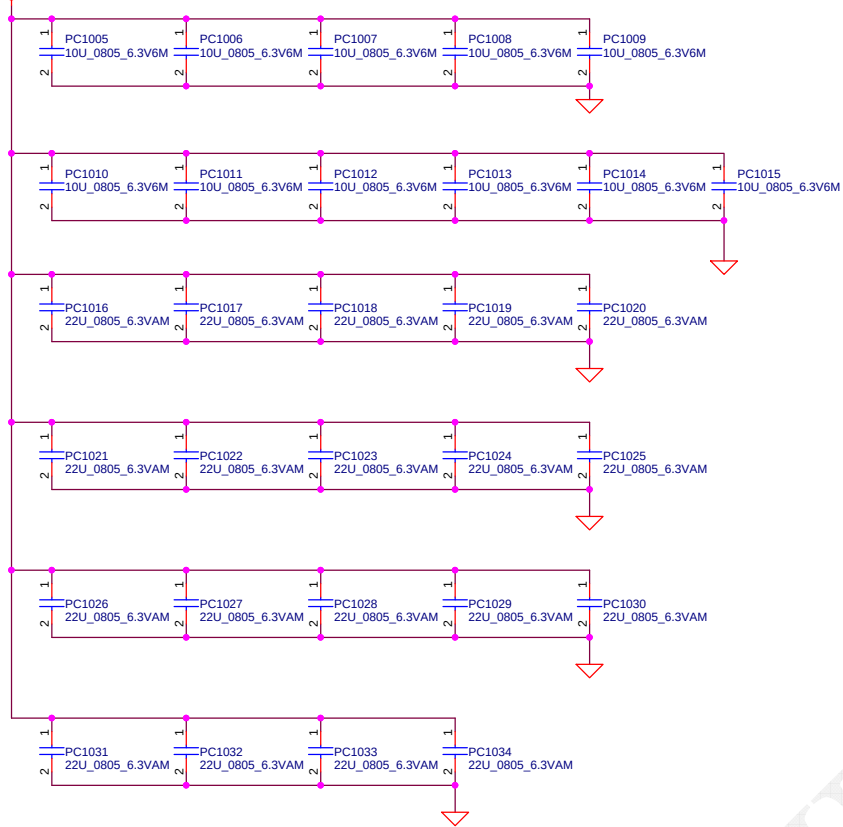
N14P-GV2 25W
Continue Current 35A
Iocp=55A-60A
PS0 Fsw=322KHZ
PS1/PS2 Fsw=475KHZ

Vboot=0.9V

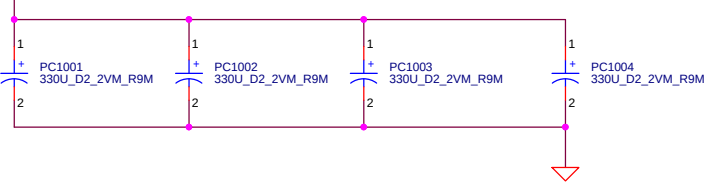
Security Classification	LC Future Center Secret Data		Title	
Issued Date	2012/12/28	Deciphered Date	2012/12/28	VGA_CORE
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+VCC_CORE



+VCC_CORE



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